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An Analysis of Risks in OTC-Derivative Pricing

Examining, quantifying, and analyzing the risks associated with the varying of a subset of configurations and input parameters using the swaption as an illustratory tool

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Abstract

The pricing of over-the-counter derivatives (OTC-derivatives) can be a complicated process, with many moving parts, where many things can go wrong. In addition, European Securities and Markets Authority (ESMA) requirements for non-cleared OTC-derivatives are somewhat non-specific. The main goal of this thesis is to perform a sensitivity analysis with respect to price on the configuration and input parameters of OTC-derivatives, with a swaption serving as the illustrative instrument. This thesis analyzes the risks of mispricing instruments from three possible sources of error; the data intraday sampling time, the model type, and the conventions used when configuring the instrument. These are compared using various risk and performance measures and discussed from an Industrial Economics and Management perspective. It is shown that the models differ in pricing most when making future predictions assuming fixed data, that the intraday data sampling time causes pricing differences sporadically, and that certain conventions exist that cause highly significant mispricings. The main conclusion from these findings is the necessity of extensive theoretical understanding when designing and auditing Mark-to-Model routines.

Keywords

OTC-derivative, OTC-instrument, Derivatives pricing, Swaptions, Swaption pricing, Organizational theory in finance, Mark-to-Model, ESMA

Sammanfattning

Prissättningen av OTC-derivat kan utgöra en invecklad process med många rörliga delar där många saker kan gå fel. Utöver det är kraven från European Securities and Markets Authority (ESMA) på icke-clearade OTC-derivat någorlunda ospecifika. Det huvudsakliga målet i detta examensarbete är utförandet av en känslighetsanalys med avseende på pris gentemot konfigurering och input-parametrar av OTC-derivat, där en swaption används som illustrativt instrument. Detta examensarbete analyserar riskerna för felprissättningar utifrån tre felkällor; datans intraday sampling-tid, modelltypen och konventionerna som används för att konfigurera instrumentet. Dessa jämförs med hjälp av olika risk/avkastnings-mått och diskuteras utifrån ett Industriell Ekonomi och Organisation-perspektiv. Det visas att modellerna skiljer sig i prissättning mest då man gör prisprediktioner samt antar fix data, att datans intraday sampling-tid orsakar skillnader i prissättning sporadiskt och slutligen att vissa konventioner finns som orsakar mycket signifikanta felprissättningar. Den huvudsakliga slutsatsen från dessa resultat är vikten av gedigen teoretisk förståelse vid design och granskning av Prissättning-efter-modell-rutiner.

Nyckelord

OTC-derivat, OTC-instrument, Prissättning av derivat, Swaptioner, Prissättning av swaptioner, Organisationslära inom finans, Prissättning-efter-modell, ESMA

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Proprietary/jointly internally developed code and AI have been used in the completion of this thesis. A usage report has been provided in section 7, the usage report and the AI usage have been approved by my adviser Kirthana Rajasekar.

The jointly developed code, along with the data provided, was used in this thesis with permission.

Disclaimer

Any opinions or motivations presented in this thesis, whether explicit, implicit, or otherwise, are not to be seen as an official standpoint or opinion of the host company, the advisers, or any other individuals involved. The thesis is an academical exploration and should only be considered as such.

Stockholm, June 2026
Viktor Gajdamowicz Bolin

Contents

1 Introduction **1**
1.1 Overview . . . 1
1.2 Purpose . . . 2
1.3 Goals . . . 2
1.4 Research Methodology . . . 3
1.5 Limitations . . . 3
1.6 Structure of the Thesis . . . 3

2 Background **5**
2.1 Relevant General Financial Theory . . . 5
2.1.1 Cleared OTC-derivatives . . . 5
2.1.2 Non-Cleared OTC-derivatives . . . 5
2.1.3 Forward Contracts . . . 6
2.1.4 Floating Rates, Indices . . . 6
2.1.5 Risk-Free Rates and Discounting . . . 7
2.1.6 Yield curves . . . 7
2.1.7 Swaps . . . 7
2.1.8 European Options . . . 9
2.1.9 Swaptions . . . 10
2.1.10 Conventions . . . 10
2.1.10.1 Business Day Conventions . . . 11
2.1.10.2 Day Count Convention . . . 11
2.1.10.3 Compounding Convention . . . 11
2.1.11 Risk/Performance Measures . . . 12
2.1.11.1 Returns . . . 12
2.1.11.2 Volatility . . . 12
2.1.12 Smiles . . . 13
2.1.12.1 The Volatility Cube . . . 13
2.1.13 The Definition of ”Mispricing” . . . 13

2.2 Relevant Financial Mathemathics 13
2.2.1 Arbitrage 14
2.2.2 Martingales 14
2.2.3 Wiener Processes 14
2.2.4 Itô's Formula 15
2.2.5 Explanation of $£^2$ 16
2.2.6 Black Scholes Formula for a Simple European Option 17
2.2.7 Expanded Black Scholes (Black-76) 19
2.2.8 Bachelier Model 19
2.2.9 Plotting Distributions 20
2.2.10 Inverting Volatilites 21
2.2.11 Linear Interpolation 22
2.2.12 Bilinear Interpolation 22
2.3 Related Work 22

3 Methodologies and Methods 23
3.1 The Specific Swaption Used 23
3.2 Research Process 24
3.2.1 Overview 24
3.2.2 Plots of NPV, Projections 25
3.2.3 Plots of Volatility 25
3.2.4 Plots of Theoretical Distributions 25
3.2.5 Varying the Convention 26
3.3 Data Collection 26
3.3.1 Assessing Reliability of Data Collected 26

4 Results 27

5 Analysis and Discussion 47
5.1 Theoretical Analysis of Results 47
5.1.1 COB/SNAP - NPV 47
5.1.2 Black-76/Bachelier - NPV 48
5.1.3 Volatility 49
5.1.4 Convention Differences at Fixing Dates 50
5.1.5 Convention Differences at Predicted Dates 51
5.1.6 Comments on Mark-to-Model 52
5.2 Applications to Industrial Economics and Management 53
5.2.1 Data 53
5.2.2 Models 53
5.2.3 Conventions 53

5.2.4 Costs 54

6 Conclusions and Future Studies **57**
6.1 Conclusions 57
6.2 Future Studies 58

7 Proprietary/Jointly Internally Developed Code, External Code Packages and AI Usage **59**
7.1 Allowed AI Usage - Course Policy 59
7.2 AI Usage Report 60
7.2.1 AI Use/Report Approval by Adviser 61

References **63**

Chapter 1

Introduction

1.1 Overview

An *OTC-derivative* is a financial agreement entered into "over the counter", meaning that it is established between two parties without the use of a regulated market (although it can be completed using central clearing). Quoting Björk, the agreement is *derivative* "[...] in the sense that it is defined in terms of some underlying asset[(s)]" [1] (p. 3). Because the contract is not traded on a regulated market, the prices are not marked-to-market but rely on models derived from financial mathematics (marked-to-model) and priced using market data. These will in theory (under mathematically and financially optimal conditions) result in a true fair value. However, due to complexities related to the real-world use of mark-to-model valuations, inconsistencies can happen. Since the global interest rate derivative OTC market turnover averages 7.9 Tn. USD per day (April 2025) [2], even small inconsistencies could result in large costs/opportunities and as such it is something worth investigating closer. Futhermore, the effects of potential mispricing are all the more relevant with regards to non-cleared physically settled OTC-derivatives as these contracts are generally priced by each side in isolation, with the only European Securities and Markets Authority (ESMA) requirements being that the margin processes are [...] robust, resilient and auditable [...]" [3] (It goes without saying that the exact requirements are slightly more extensive, but the gist is that the requirements are somewhat non-specific). As such, this thesis will focus on examining the effects on the pricing of OTC-derivatives when varying instrument configuration and input parameters, where these variations could arise from certain routines, shortcuts or operational limitations (resulting in occasional errors). The thesis will also

discuss the possible consequences and costs of the resulting inconsistencies.

The aim is to identify some especially risky or costly (or possibly profitable) aspects and give some clarity to the risks/costs/benefits of various routines. The overarching goal is to highlight certain aspects of the pricing process which might risk not being "[...] robust, resilient and auditable [...]" [3] and discussing ways this can be mitigated.

1.2 Purpose

Completing an exhaustive analysis of all ways routines might fail and the effect on OTC-derivative pricing would be an extremely extensive analysis. As such, the areas have been delimited into certain types of mispricings or routine/process failures, specified along with some added context below.

When performing the daily pricing of instruments, ideally data snapped (sampled from market) at the same time is used by both parties. The internal models of calculating the fair values might differ, however. This could result in different amounts of collateral being demanded, in turn resulting in additional costs for one party.

In the case of hedging using OTC-derivatives, the sizing of the derivatives should ideally be done using data snapped at the time of entering whatever underlying instrument is being hedged. However, it could happen that due to miscommunication, chosen incorrect routines, or technical/physical limitations, the hedging sizing is performed using data not snapped at the time of purchase. This could result in e.g. mis-sizing the hedge and gaining unwanted exposure.

Finally, the configuration of non-standard OTC-instruments in computer systems might have to be done manually (as a function of the variable nature of such contracts). Slight human mistakes could result in variations in valuation.

1.3 Goals

Since the scope of such an analysis could be very large, it has been decided to limit it to two models and two datasets, along with the misconfiguration of various common conventions in the elements underpinning the construction of an OTC-derivative. As such the goal of this thesis will be to quantify and identify the highest risk behaviors within this limited scope. The analysis, as mentioned previously, will not be fully exhaustive, but will hopefully

serve as a reminder of the costs of misconfigurations possibly attributed to behaviors/routines/processes that could be assumed as harmless.

1.4 Research Methodology

The analysis will be performed in an experimental fashion, varying the chosen factors. The analysis will be focused on varying some of the more common factors, although the exact degree of “commonness” has not been studied. In order to collect the largest number of diverse risk-factors into one instrument, ”swaptions” will be used for the analysis, more on them and other theory in Chapter 2.

1.5 Limitations

As mentioned, this analysis is not exhaustive. Any findings here should be, with reservation for any possible errors/bugs, valid. However, it must be noted that the absence of results or suggestions regarding certain routines/behaviors possibly resulting in certain changes in configuration/input and therefore effects does not necessarily mean that the routines/behaviors are absent of risks.

1.6 Structure of the Thesis

Chapter 2 presents relevant financial and mathematical theory. Chapter 3 presents the methodology of the analysis. Chapter 4 presents the results with some simpler descriptions, while Chapter 5 contains deeper analysis/discussion of the results from a theoretical viewpoint along with discussions concerning the applications to Industrial Economics and Management. Finally, Chapter 6 contains conclusions and ideas for possible future studies, and Chapter 7 presents the usage of proprietary code, external code packages and AI.

Chapter 2

Background

2.1 Relevant General Financial Theory

To understand how a swaption is priced we need to know what it is and how it is constructed. In order to do that we need to begin from the very beginning. In short, “swaptions” are “options” on “swaps”, where the “swaps” in our case will have a “floating” and a “fixed” leg. The floating leg relies on “floating rates”, which are sampled from a “yield curve”, consisting of “forward” rates. Starting from the bottom, and building upwards, this jargon will be given context in the following section.

2.1.1 Cleared OTC-derivatives

Cleared derivatives are centrally priced (the Central Counterparties (CCP) demands margin from members, [4]). This clearing is initated through a process called “novation” where “The CCP becomes the buyer to the seller and the seller to the buyer in the original contract” (p. 5) [4]. As such the analysis can be relevant in situations where we value connected positions internally (such as hedging the not-yet-cleared OTC-derivative), and thereby one could in theory risk mis-sizing, for example, the hedge.

2.1.2 Non-Cleared OTC-derivatives

In non-cleared derivatives, there is no central entity providing regular prices and so the two actors rely on, subject to ESMA requirements [3] and whatever bilateral agreement might be in place, their own models. Because there now exists counterparty-risk, the parties (generally) request collateral, based on

their own models. Often the pricing results are only compared in the case where amounts of collateral outside of some margin of error is requested, so it is the case (and was found in the somewhat related article "Closing Time Effects on Derivative Pricing and Risk Measurement" by Andreas Melin [5]) that one party can get systematically disadvantaged by "[...] paying significantly more collateral than [they] statistically should" (p. 51) [5] (whether intentionally or not).

2.1.3 Forward Contracts

Quoting Chapter 2 in A Primer for Financial Engineering, "A forward contract is an agreement between a seller and a buyer to deliver and purchase, respectively, a particular amount of an asset, an underlying, at a predefined price (delivery price) at a predefined date, delivery date." [6]

In other words, forward contracts, also known as "forwards", are financial contracts that give the *obligation* to enter into some transaction at a specified future date. Forwards or rates are what constitute the "yield curve", which is something we will need to price floating rate coupons.

2.1.4 Floating Rates, Indices

It is commonly known that when we take a loan from a bank, the loan in general has a cost. We can either choose to pay this cost in fixed installments or floating installments. The fixed payment is defined by some fixed rate (rate of return) representing some percentage of the initial loan which we will have to pay above just returning the nominal loan. The floating payment on the other hand is defined directly or indirectly by a benchmark, an "index", which for our purposes can be described as a floating rate published by some institution responsible for the index, often based on some public fixed methodology. When entering into loans on open financial markets they are for the most part defined on indices with public methodologies, which is very helpful in predicting future floating payments/rates, which in turn is vital to estimating/calculating the total cost of a floating loan.

In our swaption we will use the 3-month STIBOR (Stockholm Interbank Offered Rate) [7], [8] (published by the Swedish Financial Benchmark Facility (SFBF)) and the SWESTR (Swedish krona Short Term Rate), published by Riksbanken [9], [10].

2.1.5 Risk-Free Rates and Discounting

If we derive such rates and yield curves from an index based on the cost for some underlying loan with no risk of default, we receive a "risk-free rate" (RFR). Since this can be considered a guaranteed return on an investment it can be used to discount future values/cash flows, ergo the formula for converting mediums of exchange between two points in time is given by:

$$X_t \cdot (1 + RFR_{t \to T}) = X_T \tag{2.1}$$

Where X_t and X_T are nominal values at times t and T, and $RFR_{t \to T}$ is the cumulative risk-free rate over the period $[t, T]$. If we want to discount a certain future nominal value at time T to the present time t, we simply solve for X_t.

2.1.6 Yield curves

In order to price floating rate-based contracts we need to know what the floating rate is expected to be in the future. "yield curves" are a collection of "maturities" (length of time of a loan, or the time until repayment) and the expected interest rate over that period ([11], p. 50), and they will be used to generate floating coupons and discount future payments to the present value.

Yield curves are constructed using forwards (synthetic forwards). If we define a forward for a simple contract connected to the a floating rate defined by some specific index we can through the concept of "risk-free arbitrage" ([11] p. 32) use current rates to get a fair prices for this contract, implying a certain objective expected rate in the future. ([11] p. 35). If we then plot the rates of these contracts for different maturities, we receive the yield curve.

2.1.7 Swaps

A swap can in short be described as a bilateral agreement with the purpose of exchanging two streams of cash flows (based on [11], p. 144). These cash flows can in theory be of any type, and the swaps are named accordingly. One of the most common swaps, an interest rate swap (IRS), is a swap where the flows differ in coupon base (e.g. a floating rate coupon being exchanged for a fixed rate coupon). The cash flows have multiple properties, such as business day conventions, convention of calculating the payments/coupons (day counter conventions) and payment frequency. The swap contract also has a defined duration, called the contract's "tenor". A very common version is the fixed/floating swap, and "swaptions" based on these IRS-es will be the

main focus in this thesis.

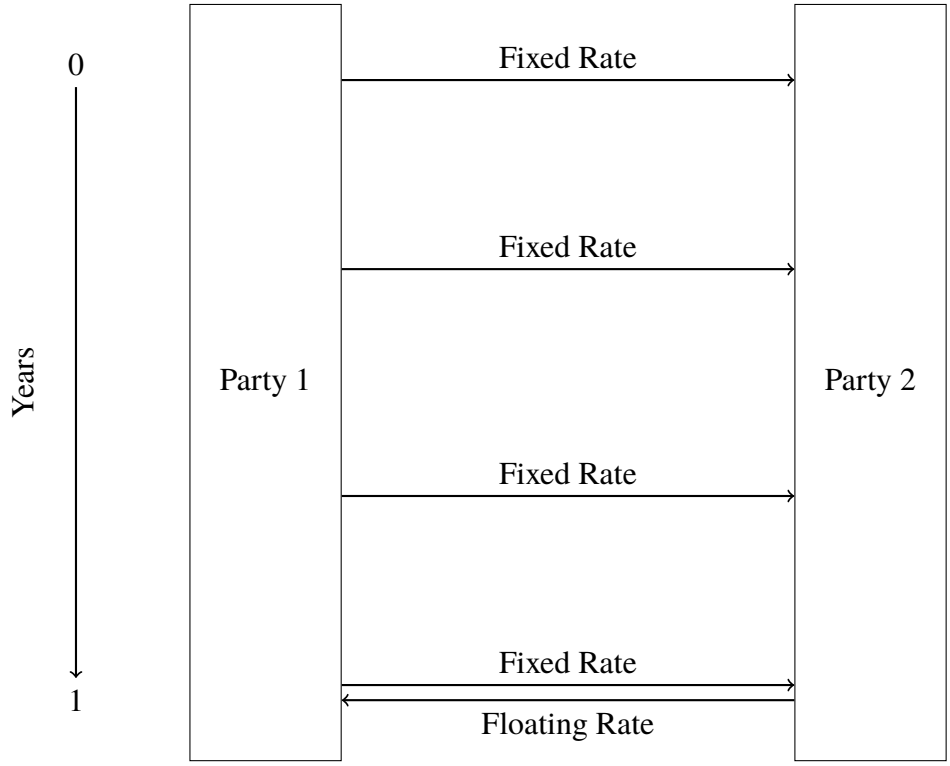


Figure 2.1: Cash flows of a fixed/floating swap with tenor 1Y and no exchange of nominal. Fixed payments have a quarterly frequency while the floating is annual.

In order to value an IRS, we need to know the present value of all the coupons. We use the discounting formula mentioned previously, taking the RFR for each point in time from the relevant yield curve, solving for the present value of all individual cash flows. We then get the IRS Net Present Value (NPV) by adding all incoming discounted flows and subtracting all outgoing. If we call the fixed nominal payment X_{fixed} and the floating nominal payments X_{T_j} where $T_i \in \{Fixed\ payment\ dates\}$ and $T_j \in \{Floating\ payment\ dates\}$ and $RFR_{t \to T}$ is the cumulative risk-free rate over the period $[t, T]$ we get the NPV at time t as:

$$NPV_{IRSReceiver}(t) = \sum_{T_i} (X_{fixed} \cdot \frac{1}{(1 + RFR_{t \to T_i})}) - \sum_{T_j} (X_{T_j} \cdot \frac{1}{(1 + RFR_{t \to T_j})}) \quad (2.2)$$

(Formula based on [11], Chapter 9)

Here the "receiver" in the subscript is signifying that we are receiving fixed

and paying floating. Multiply by -1 to get the NPV of an IRS where the floating payments are being received.

2.1.8 European Options

Options are a type of instrument where the owner has the right to purchase or sell the underlying instrument (exercising the option) at a certain price, known as the "strike" price, during some defined time or at some defined date. Options are a type of "contingent claim", which is (According to [1] (p.3)) due to "[...] the value of the option is contingent on the evolution [in time] of [an aspect of underlying]". But this would not differentiate a contingent claim from, say, forward contracts (the value of which also depends on the evolution of the underlying). The main difference in my view is therefore that the claim ("exercise") is *optional* in a contingent claim, meaning that the claim itself is "contingent" on some evolution of events.

The value of an option k at expiry Π_T^k, where T is the expiration is given by the following formula:

$$\Pi_T^k = \max(0, V_u(T) - S_k) \tag{2.3}$$

(Formula based on [1])

Here $V_u(T)$ describes the value of the "price evolution" of the underlying u at time T and S_k is the strike price. For valuing the option at times before expiry we will require more complex formulas, which will be explained in the coming sections. Options that can only be exercised at the time of expiry are called "European".

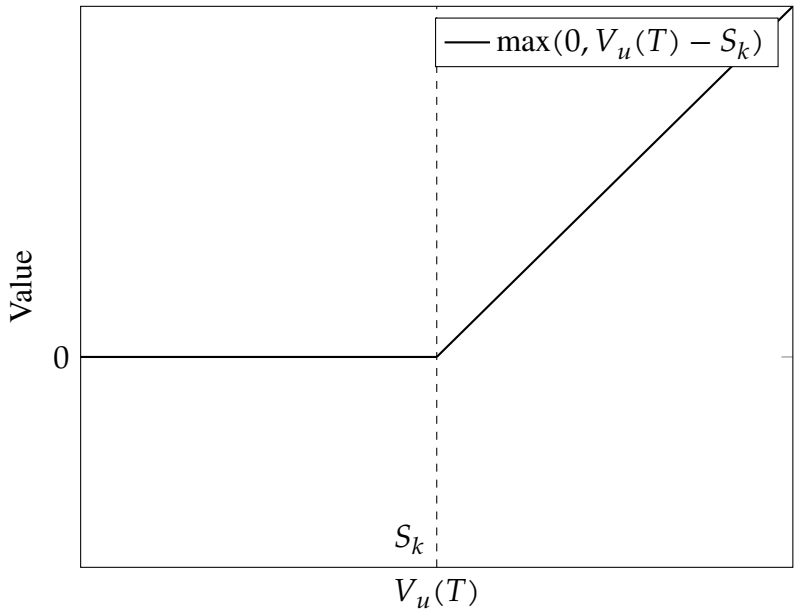


Figure 2.2: The value Π_T^k of a European option k with strike S_k against the underlying value $V_u(T)$ at time T (expiry).

2.1.9 Swaptions

Swaptions are options on swaps [11] (p. 368), where the strike is some fixed rate connected to the underlying IRS. They are commonly differentiated according to who sends or receives the fixed rate. If the underlying swap gives the option purchaser a right to receive a fixed cash flow, it is called a "receiver" swaption. Conversely, it is called a "payer" swaption if the underlying swap gives the buyer of the option the right to pay a fixed rate [12] (p. 298).

The value of a swaption with European exercise follows a similar formula to the "standard" European option (as in a "normal" stock as underlying), where the value of the swaption depends on the value of the underlying swap. In the case of valuing the swaption at exercise the formula is identical. As mentioned above, the pricing formulas for pricing at points in time before expiry are more complicated and will be explained in the coming sections.

2.1.10 Conventions

One of the main complications of pricing swaptions are the *conventions*. I will focus on the *Business Day Convention*, *Day Count Convention* and *Compounding Convention*. Conventions add a large degree of complexity

when attempting to accurately price instruments with many "moving parts". Often these conventions are specified for each sub-instrument (swap legs, the underlying swap, the swaption itself, indices connected to the legs). These can co-interact and overlap with each other. In specifying one swaption there can easily be 10-15 unique conventions to keep in mind, not including co-interaction conventions and the conventions on the data itself.

2.1.10.1 Business Day Conventions

Quoting the QuantLib documentation [13]:

- Following: "The date is corrected to the first working day that follows."
- ModifiedFollowing: "The date is corrected to the first working day after that, unless this working day is in the next month; if the modified working day is in the next month, the date is corrected to the last working day that appears before, to ensure the original The date and the revised date are in the same month."
- Preceding: "Correct the date to the last business day that Preceding before."
- ModifiedPreceding: "modify the date to the last working day that appeared before, unless the working sunrise is now the previous month; if the modified working sunrise is now the previous month, the date is modified to the first working day after that The original date and the revised date are guaranteed to be in the same month."

2.1.10.2 Day Count Convention

The day count convention defines interest accrual, and as such the coupon payments and NPV of fixed income instruments. Some of these can get quite complicated. For a more or less complete list I refer to the 2006 ISDA Definitions [14].

The simplest conventions, however, are the ACT/365 (Fixed) and ACT/360. They divide the actual number of days in the compounding period (such as a quarter) by 365 and 360 respectively [14] (p. 11).

2.1.10.3 Compounding Convention

From the QuantLib source code [15] we can see that the compounding conventions are defined as follows (quoting the formulas/definitions):

- Simple -> "$1 + rt$"
- Compounded -> "$(1 + r)^t$"
- Continuous -> "e^{rt}"
- SimpleThenCompounded -> "Simple up to the first period then Compounded"
- CompoundedThenSimple -> "Compounded up to the first period then Simple"

where r is denoting interest rate, and t time.

2.1.11 Risk/Performance Measures

2.1.11.1 Returns

Returns can be presented either as absolute values, or as a percentage change.

$$r_{abs} = V_{t+n} - V_t \tag{2.4}$$

$$r_{rel} = \frac{V_{t+n} - V_t}{V_t} \tag{2.5}$$

Where r denotes our returns, V_t the value at time t and n being our rolling window size.

2.1.11.2 Volatility

The volatility σ of an instrument is given by the standard deviation of its returns [16].

By considering the return R_i can be over any defined period length t, we get the following formula for $i \in \{1, \ldots, n\}$ data points over some total time $T = nt$:

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (R_i - \bar{R}_i)^2} \tag{2.6}$$

Where

$$\bar{R}_i = \frac{\sum_{i=1}^{n} R_i}{n} \tag{2.7}$$

As with the standard deviation in general, the volatility returns the spread of returns within one standard deviation of the mean. As such it is useful for quantifying the size of rare market movements.

2.1.12 Smiles

When pricing options we sample implied volatility from the market by solving for the volatility that solves a given pricing model. If we then plot the volatility against the strike, we gain a “volatility smile”. In theory these should be flat (independent of, among other things, the swaption strike. This is a model assumption, see [17]), but in practice the volatilites connected to a strike shifted relative to the ATM value are generally slightly higher, resulting in the so called “smile” [17].

2.1.12.1 The Volatility Cube

If we take such smiles and plot them for each swap tenor (meaning contract length) and swaption time to maturity, we get a volatility cube. This is used to sample a volatility value for use in pricing an instrument, either by some sort of interpolation or by a parametric solution.

These are generally quoted as normal (Bachelier) volatilites [18], more on this further down.

2.1.13 The Definition of ”Mispricing”

Since marking-to-model lacks a ”true” price in the mark-to-market sense, the use of ”mispricing” in this text should be interpreted as the *price-misalignment* relative to the relevant contextual baseline.

2.2 Relevant Financial Mathemathics

The following section is not strictly neccecary for the reader happy with accepting that the marking-to-model is fair/real (given some varying assumptions), even though they are not directly sampled from the market. Still, I wanted to at least include something to serve as a high-level overview and context for the discussion and conclusions, or as a simple introduction for the interested reader.

Now an extremely simplified overview of the derivation of the pricing formulas will be presented (Itô Formula and Black-Scholes).

The below section is based on (Björk, [1]) if not mentioned otherwise, presented in a "loose" manner ommitting many steps and exact definitions for brevity. It is also assumed that the reader has a general understanding of stochastics/statistics and differential equations.

2.2.1 Arbitrage

One of the fundamental assumptions underpinning the unique, fair pricing of options is that the market is free of arbitrage. Arbitrage is defined by the following equations holding for a "self-financed" portfolio h (i.e. formed by loaning and buying on the market with no "free" cash) with value $V^h(t)$ at time t [1] (p. 96):

$$V^h(0) = 0 \tag{2.8}$$

$$P(V^h(T) \geq 0) = 1 \tag{2.9}$$

$$P(V^h(T) > 0) > 0 \tag{2.10}$$

An arbitrage market would mean that there is no portfolio that can be constructed such that a profit is guaranteed without possibility of losses. This underpins constructing "fair" prices for various contingent claims.

2.2.2 Martingales

The main point of martingales in this narrow context is that the price process of any asset following certain conditions can be "normalized" into a martingale, resulting in a process with an expected "profit" of 0 at all times t, i.e. a "fair" process/game. This normalization is connected to the so-called **Risk Neutral Valuation**, which is the arbitrage free pricing of an instrument. It can be shown that it is only possible when the market is free of arbitrage, and that happens iff a "martingale measure Q" exists, under which we can calculate these fair prices in question. Very generally put, calculations under the measure Q correspond to isolating the deterministic part (drift) of a price, leaving only a transformed ($W(t)$ vs $\bar{W}(t)$) stochastic (random/"noisy") part. The following parts will add more context, and it will hopefully become a bit clearer how the measure Q manifests in practice.

2.2.3 Wiener Processes

Paraphrasing/quoting from (Björk, [1]), a stochastic process W is a Wiener Process if:

- $W(0) = 0$
- The process W has independent increments, meaning given disjoint time segments, the processes W over those are independent stochastic variables
- Given s < t, $W(t) - W(s) \sim N(0, t-s)$, $N(\mu, \sigma^2)$ signifying a normal distribution with mean μ and standard deviation σ
- W has continuous trajectories

Now we use this to form an approximation of the stochastic diffusion process X

$$X(t + \Delta t) - X(t) = \mu(t, X(t))\Delta t + \sigma(t, X(t))\Delta W(t) \tag{2.11}$$

Where

$$\Delta W(t) = W(t + \Delta t) - W(t) \tag{2.12}$$

In the Wiener process, μ is a "locally deterministic drift", while σ is the term representing a normally distributed disturbance or. Extrapolating to finance, μ would be (something like) some locally deterministic long-term drift/trend in the price evolution, while σ is the local stochastic "uncertainty/randomness".

One would want to solve this by some version of dividing by letting $\Delta t \to 0$ and integrating over the trajectory of W, or by dividing first and thereby deriving by time. Both of these are not formally possible, the derivation since it can be shown that W is undifferentiable everywhere and the integral since it can be shown that the infinitesimal "sub-trajectories" of W have a locally unbounded variation (Both direct reference to Björk Chapter 4 [1]). As such we need a solution, which comes in the form of Itô integrals.

2.2.4 Itô's Formula

We consider instead the integral mentioned before in a more global sense. It can be shown that it is possible to have an L^2 (square integrable) integral of the same form, where the stochastic process $g(s) \in £^2$ (defined below):

$$\int_0^t g(s)dW(s) \tag{2.13}$$

This is called the Itô integral, and it is used to derive the Itô calculus which is used to solve the underlying integral for all $t > 0$ (gained from letting $t \to 0$

as mentioned above):

$$X(t) = a + \int_0^t \mu(s)ds \; + \; \int \sigma(s)dW(s) \tag{2.14}$$

This equation is fundamentally what is being used when modeling/describing/pricing various financial instruments through, among other formulas, the derived 1976 Black-Scholes (reference) formula and various extensions of it. It can also be written as a more intuitive "stochastic differential", although the "real" equation is still the integral.

$$\begin{cases} dX(t) = \mu(t, X(t))dt + \sigma(t, X(t))dW(t), \\ X(0) = a \end{cases} \tag{2.15}$$

Since X models the underlying, we will want to know the price of some *derivative* of X, $Z(t) = f(t, X(t))$. Like with all other omitted steps, I refer the curious reader to Björk (2009) [1]. At the end, however, we arrive at the Itô formula, here presented in differential form with shorthand notation, omitting variable dependence and $\frac{\partial}{\partial x}$ being the derivative w.r.t. X:

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX)^2 \tag{2.16}$$

With the following rules:

$$\begin{cases} (dt)^2 = 0, \\ dt \cdot dW = 0, \\ (dW)^2 = dt \end{cases} \tag{2.17}$$

2.2.5 Explanation of $\pounds^2$

When one wants to formulate the Itô integral as a stochastic integral we need to define the class $\pounds^2$ mentioned above. Quoting/paraphrasing (Björk, [1]), we have that:

The stochastic process $g \in \pounds^2[a, b]$ if

1. $\int_a^b E[g^2(s)]ds < \infty$
2. g is adapted to the $\mathcal{F}_t^W$−filtration ($g(t) \in \mathcal{F}_t^W$)

And

- $g \in \pounds^2$ if $g \in \pounds^2[0, t] \; \forall \; t > 0$

$E[\,\cdot\,]$ denoting the expected value. Omitting further steps, it can be shown that if $g \in \pounds^2[a,b]$, $\int_a^b g(s)dW(s)$ is $\mathcal{F}_b^W$ **measurable**. Ommitting steps once again, the main point of this result is that all stochastic integrals on this form are in fact Martingales, which can be said seems reasonable since they stem from a "fair" Wiener-process (no drift)

For the definition of filtration/measurability, please consult your literature of choice on measure theory or see [1] (p. 43).

2.2.6 Black Scholes Formula for a Simple European Option

We start with two assets (Note fixed σ, α, r, and the bar on top of W):

$$\begin{cases} dB(t) = rB(t)dt \\ dS(t) = \alpha S(t)dt + \sigma S(t)d\bar{W} \end{cases} \tag{2.18}$$

B being the price process for a risk-free bond and S the price process for some simple instrument. The reason we have multiplied the wiener process by $S(t)$ is because we assume that both the local drift and volatility are proportional to the price. This is also known as assuming a log-normal price distribution and is a fundamental assumption in the Black-Scholes pricing model.

In short, we define the price function of the derivative instrument $\Phi(s)$, apply the Itô formula and create a portfolio of the underlying and derivative, and simplify. Using the fact that a self-financing portfolio with no arbitrage possibility restricts this relative portfolio, we substitute in r according to this relation and receive the Black-Scholes Equation in a PDE-form (where F is a price process of the derivative, see [1] (p. 99)):

$$\begin{cases} F_t(t,s) + rsF_s(t,s) + \dfrac{1}{2}s^2\sigma^2 F_{ss}(t,s) - rF(t,0) = 0 \\ F(T,s) = \Phi(s) \end{cases} \tag{2.19}$$

With the final equation being equivalent to the option price at expiry mentioned in the earlier theoretical segments. This PDE-system is solved by the "Feynman-Kač stochastic representation formula" (Björk, [1] p.73) (will not be derived here)

$$F(t,s) = e^{-r(T-t)}E^{t,s}[\Phi(X(T))] \tag{2.20}$$

Where X follows

$$\begin{cases} dX(u) = rX(u)du + X(u)\sigma(y, X(u))dW(u) \\ X(t) = s \end{cases} \tag{2.21}$$

[1] (p. 102). This stochastic process X where the stochastic process is on Q measure, compared to S previously being defined on whats called P measure. This is the explanation for the use of bar on W, as $\bar{W}$ is the P-Wiener Process, while W is a Q-Wiener Process.

In summary, the deriving of the price equations come from a P measure, but the Feynman-Kac solution to the price equations is given over the expectation on Q (with rate r, so Q measure in the self financed portfolio sense). In essence this implies the fact that all instruments have the same expected drift (RFR), while only differing in risk level. As such we use the expected value over Q as we are taking the expected value of the Q-measure process of the same form (the X-process). This expected value is denoted by $E^Q_{t,s}[\Phi(S(T))]$, which symbolizes the conversion of S into the RFR-form of X and taking the expected value of that.

If the mention of a Q-measure in the martingale section felt unclear, hopefully this clears it up a bit! It is basically a transform of a pricing dynamic into a "fair" price process, "fair" being used here in the same way one might in describing a "fair" game (no expected profit) in the context of an arbitrage free market with a risk-free rate r.

Finally, we can now write out the price formula:

$$F(t,s) = e^{-r(T-t)}E^Q_{t,s}[\Phi(S(T))] \tag{2.22}$$

Which is solved in the case of a European call (right to buy) option with strike K and expiry at time T where $t \in [0,T]$:

$$F(t,s) = sF_{N(0,1)}(d_1(t,s)) - e^{-r(T-t)}KF_{N(0,1)}(d_2(t,s)) \tag{2.23}$$

Where

$$F_{N(0,1)}(x) = \int_{-\infty}^{x} f_{N(0,1)}(y)dy \tag{2.24}$$

$$d_1(t,s) = \frac{ln(\frac{s}{K}) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \tag{2.25}$$

$$d_2(t,s) = d_1(t,s) - \sigma\sqrt{T-t} \tag{2.26}$$

and $f_{N(0,1)}(x)$ is the density function for the $N(0,1)$-distribution (so

$F_{N(0,1)}(x)$ is the cumulative distribution function for the $N(0,1)$-distribution).

2.2.7 Expanded Black Scholes (Black-76)

The Black-Scholes equations can also be solved for other underlying instruments. In "The pricing of commodity contracts" from 1976, Fischer Black derives the following formula relevant in use with rate contracts. [19] We will omit the complete derivation.

$$\Pi_{payer}(s,t) = e^{r(t-T)}\left[sF_{N(0,1)}(d_1) - KF_{N(0,1)}(d_2)\right] \tag{2.27}$$

$$\Pi_{receiver}(s,t) = e^{r(t-T)}\left[KF_{N(0,1)}(-d_2) - sF_{N(0,1)}(-d_1)\right] \tag{2.28}$$

$$d_1 = \frac{ln(\frac{s}{K}) + \frac{\sigma_B^2}{2}(T-t)}{\sigma_B\sqrt{(T-t)}} \tag{2.29}$$

$$d_2 = \frac{ln(\frac{s}{K}) - \frac{\sigma_B^2}{2}(T-t)}{\sigma_B\sqrt{(T-t)}} \tag{2.30}$$

Where s is the forward swap rate (i.e. the rate that results in a fair price of a forward on the underlying swap (NPV = 0), basically the rate that gives the underlying a fair price), K the strike rate, r the RFR, T the time of expiry and t the current time.

2.2.8 Bachelier Model

The Bachelier pricing model is similar to the Black-Scholes model, but it assumes normally distributed returns instead of log-normal. It is derived from the following sets of assets:

$$\begin{cases} dB(t) = rB(t)dt \\ dS(t) = \alpha dt + \sigma_N dW \end{cases} \tag{2.31}$$

Which as mentioned previously implies Normal returns. The resulting formula has the form (from [18])

$$\Pi_{payer}(K,t) = e^{r(t-T)}\left[(s-K)F_{N(0,1)}(d_N) + \sigma_N\sqrt{(T-t)}f_{N(0,1)}(d_N)\right] \tag{2.32}$$

$$\Pi_{receiver}(K,t) = e^{r(t-T)}\left[(K-s)F_{N(0,1)}(-d_N) + \sigma_N\sqrt{(T-t)}f_{N(0,1)}(d_N)\right] \tag{2.33}$$

$$d_N = \frac{s - K}{\sigma_N \sqrt{(T - t)}} \tag{2.34}$$

Note that $T - t$ is equal to the time to maturity.

2.2.9 Plotting Distributions

In order to interpret differences in model pricing of swaptions it can be useful to see what the probability density distributions for the underlying look like for the two models. General lognormal $LogN(\mu, \sigma^2)$ and normal $(N(\mu, \sigma^2))$ distributions follow the following forms [20] [21]:

$$f_{LogN}(x) = \frac{1}{x\sigma\sqrt{2\pi}} exp\left\{-\frac{(ln(x) - \mu)^2}{2\sigma^2}\right\} \tag{2.35}$$

$$f_N(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left\{-\frac{(x - \mu)^2}{2\sigma^2}\right\} \tag{2.36}$$

We need μ, σ for our underlying swap given our model assumptions.

For Black-76, we begin by considering the swap with the dynamics given by the following equation:

$$\begin{cases} dS(t) = \sigma S(t) dW^Q(t), \\ S(0) = s \end{cases} \tag{2.37}$$

Björk (Chapter 7.5) [1] gives (modified for our assumptions and writing W explicitly as a Q-Wiener Process (as is done in the 4th edition [22])):

$$\begin{aligned} S(T) &= s \cdot exp\left\{-\frac{\sigma^2}{2}(T - t) + \sigma(W^Q(T) - W^Q(t)\right\} \\ &= s \cdot exp\left\{-\frac{\sigma^2}{2}(T - t) + N(0, \sigma^2(T - t))\right\} \end{aligned} \tag{2.38}$$

where the second equality was derived using the definition of the Wiener Process, along with allowing for some abuse of notation.

Since this underlying asset follows a log-normal price movement, we can get the expected value μ and σ using the relation

$$N(\mu, \sigma^2) = ln(LogN(\mu, \sigma^2)) \tag{2.39}$$

where μ, σ correspond to the expected value and standard deviation of the

corresponding normal distribution (following the convention used in the beginning of this section [21]). We get this by taking the natural logarithm:

$$\begin{aligned} ln(S(T)) &= ln(s) + -\frac{\sigma^2}{2}(T-t) + N(0, \sigma^2(T-t)) \\ &= N(ln(s) - \frac{\sigma^2}{2}(T-t), \sigma^2(T-t)) \end{aligned} \tag{2.40}$$

$$\Rightarrow$$

$$\begin{aligned} \mu &= ln(s) - \frac{\sigma^2}{2}(T-t) \\ \sigma_{scaled} &= \sigma\sqrt{(T-t)} \end{aligned} \tag{2.41}$$

For Bachelier the normal swap dynamics as given in 2.2.8 (including $\alpha = 0$ by assumption) are used. The swap price process is normally distributed and as such centered on the forward swap rate ($\mu = s$). Omitting exact steps for brevity, it can be shown that the standard deviation also includes the same time-to-maturity-adjustment as the Black-76 model, $\sigma_{scaled} = \sigma\sqrt{(T-t)}$. As a pseudo-motivation, one can look at the denominator in the d_1, d_2, d_N-terms, which all contain $\sigma\sqrt{T-t}$.

Entering the corresponding expected values into the general formulas for the distributions we get the following formulas:

$$f_{Black76}(x) = \frac{1}{x\sigma\sqrt{(T-t)}\sqrt{2\pi}} exp\left\{-\frac{(ln(x) - ln(s) + \frac{1}{2}\sigma^2(T-t))^2}{2\sigma^2(T-t)}\right\} \tag{2.42}$$

$$f_{Bachelier}(x) = \frac{1}{\sigma\sqrt{(T-t)}\sqrt{2\pi}} exp\left\{-\frac{(x-s)^2}{2\sigma^2(T-t)}\right\} \tag{2.43}$$

where the σ are the corresponding volatilities for the respective formulas mentioned earlier.

2.2.10 Inverting Volatilites

Since our data only contains the normal volatilites, we need to invert these (into log-normal volatilities). This is done by calculating the Bachelier price and saving the strike and forward rate. Then the Black-76 formula is solved for the log-normal volatilites after equating it to the Bachelier price. These then are the arbitrage free market implied log-normal volatilites.

2.2.11 Linear Interpolation

Linear interpolation is the simplest interpolation alternative and commonly used. Also common is cubic splines, but we will not use them in our analysis. Linear interpolation is here used in both interpolation of yield curves, but also of the volatility cube [23].

Given two points (x_0, y_0) and (x_1, y_1), and given $x_0 < x < x_1$ we linearly interpolate these points (giving us a straight line y) by [24]

$$y = \frac{(y_0(x_1 - x) + y_1(x - x_0))}{(x_1 - x_0)} \tag{2.44}$$

2.2.12 Bilinear Interpolation

Bilinear interpolation is also used in the interpolation of the volatility cubes [23].

In bilinear interpolation we fit a "hyperbolic paraboloid" $f(x, y)$ through four adjascent points [25].

$$f(x, y) = ax + by + cxy + d \tag{2.45}$$

Where

$$f(x, y) = (f_{10} - f_{00})x + (f_{01} - f_{00})y + (f_{11} + f_{00} - f_{01} - f_{10})xy + f_{00} \tag{2.46}$$

And $f_{mn} = f(m, n)$ denote the adjacent points.

2.3 Related Work

The main similar work found is a Master thesis called "Closing Time Effects on Derivative Pricing and Risk Measurement" (2014) by Andreas Melin [5], which focuses on the effect of the choice of "daily return intervals", compared to this thesis which focuses on the effects of a certain subset of model configurations and input parameters.

Chapter 3

Methodologies and Methods

As mentioned before, swaptions will be used as the instrument under examination because of their complexity. Swaption pricing is influenced by interest rate curves, curve volatility and more, in addition to having many ”moving parts” when configuring it. It reflects many market aspects and should give a good illustration of the total risks.

3.1 The Specific Swaption Used

10Y tenor swap, with a $\approx$ 1 year maturity from 20th Jan 2026, one leg is quarterly payment STIBOR with 3-month maturity, and the other leg is fixed 2.5% yearly. The swaption is a receiver. The sub-instruments are constructed as follows:

- Discounting curve (SWESTR): Act/360, Linearly interpolated, Annual frequency, Compounded
- Floating curve (STIB3M): Act/365 (Fixed), Linearly interpolated, Annual frequency, Compounded
- Floating index: 2 days spot lag, ModifiedFollowing, No EOM adjustment, Act/360
- Volatility cube ATM surface: Following, Act/360
- Fixed leg: ModifiedFollowing
- Floating leg: ModifiedFollowing

- Underlying swap: Fixed daycount: 30E/360, Floating day count: Act/360

The nominal value is set to 1, this results in the NPV of the swaption representing the share of the nominal that it is worth (and as such is unit-less).

With this configuration our swaption is out of the money for the entire data range, which will play a key role in the exact behavior of the results and their interpretation.

3.2 Research Process

The analysis will consist of time series analysis over the various combinations of financial models, data and random errors mentioned, using the above-mentioned risk and performance measures. The possible causes/underlying mechanics will be discussed and connected to organizational routines from a financial industry/organizational perspective.

3.2.1 Overview

For the floating curve, two datasets used will be used. One "snapped" at 15:30 (henceforth SNAP), and another sampled at the closing auction at 17:30 ("close of business", henceforth COB). The data has been sampled by SEB Asset Management; one financial quarters worth will be used in the analysis. All other data (volatilites, discounting curve) will be all at COB.

In order to collect the largest number of diverse risk-factors into one instrument, it has been decided to use "swaptions" for the analysis.

The two pricing models chosen are:

1. the "Black-76" swaption pricing formula [19] (pricing of options on futures)

2. the Bachelier pricing formula [26]

These were chosen because of how common they are, and because while similar, they should vary in theoretically predictable ways (notably, the Bachelier formula allows negative prices (or rates in our case)).

The conventions will be varied in different sub-instruments of the swaption.

The python package "QuantLib" [27] will be used to construct and price the swaption and all sub-components constituting it.

Prices will be calculated for all available dates, along with predicted forward in time from the given dates, for various combinations of models, data and incorrect conventions in various sub-instruments. On these time-series, visualisation and quantification of mispricings will be performed. Various risk/performance measures (such as volatility and NPV-time series) will also be calculated and visualized.

Finally, these results will be discussed and actionable insights with regards to pricing-process/system design will be presented.

3.2.2 Plots of NPV, Projections

The most basic analysis consists of examining the "raw" NPV. This is also what determines the pricing differences. The NPV can be calculated for each "fixing day" (day that data exists). Differences here illustrate the same day mispricing of an instrument due to whichever factor we chose to vary.

The NPV can also be calculated as a projected value in the future by increasing t (reducing the time-to-maturity (TTM)) in the pricing formulas. This axis will be called "Days out" in the results. Differences here can be relevant in the case of planning purchases/hedges in the future, such as planning a purchase of a swaption x days out.

Note that this is essentially "Theta", defined as "[...] the change of premium for a unit change in the time to maturity [...]" ([11], p. 315), because this simple TTM-shift assumes all other inputs (mainly volatility) are constant. As such the exact value is not necessarily an arbitrage free price, but it does show the time dependence of the swaption in isolation.

3.2.3 Plots of Volatility

In order to get a glimpse of the underlying mechanics we will also plot the volatility of projections various days out. It will allow us to get an understanding of the absolute and relative distribution behaviors of the various permutations when increasing the "Days out".

3.2.4 Plots of Theoretical Distributions

The distributions of the underlying swap at any fixing date and any "days out" are analytical, and we will plot these to better understand the dynamics of pricing differences between the models. They follow section 2.2.9.

3.2.5 Varying the Convention

All conventions will be varied one at a time. The floating curve will vary the Compounding Convention and the Day Count Convention. The fixed leg in the swaption will vary the Business Day Convention and the Day Count Convention.

3.3 Data Collection

A quarters worth of daily historic data (time series) for the Normal (for Bachelier) implied volatility cube, discounting curve (SWESTR) and floating curve (STIBOR3M) at COB has been provided by SEB Asset Management. SNAP data is also available for STIBOR3M. Data is market data gathered by SEB Asset Management.

3.3.1 Assessing Reliability of Data Collected

While the data is reliable, the interpretation is more difficult. Large amounts of verification have been done in order to ascertain the correct conventions, but as will be seen some smaller issues seem to remain. The risk of additional issues/misconfigurations not yet discovered is not nil.

Chapter 4

Results

The following plots contain results from the previously described methodology, along with some context describing them. More in-depth interpretations/discussions will be located in the appropriately named subsequent section.

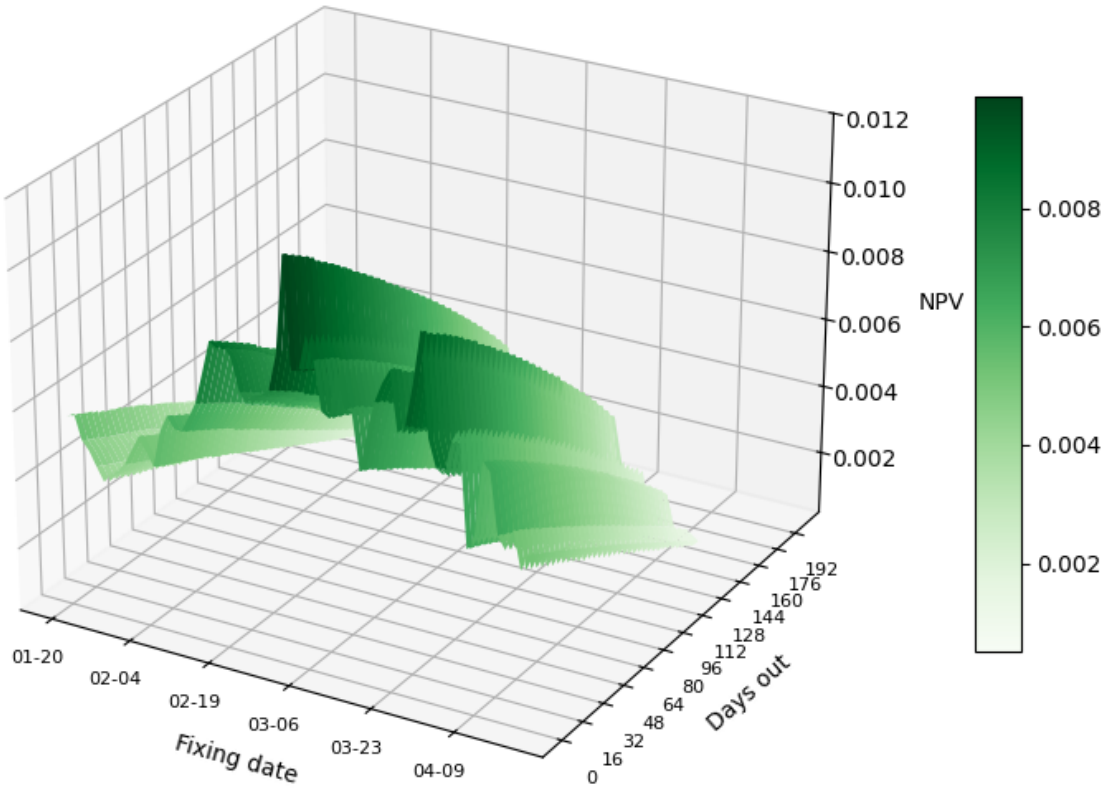


Figure 4.1: Bachelier pricing surface with COB curve data. Days out signify reduced time to maturity for the swaption (predicted price) from fixing date and visualize the natural price decay as the option is out of the money.

This figure describes the pricing surface of our swaption. The line traced nearest the fixing date axis is the price at 0 days out, pricing at fixing date. Traveling out on the Days out axis we can see the natural decay of the NPV as the time to maturity $T - t$ decreases.

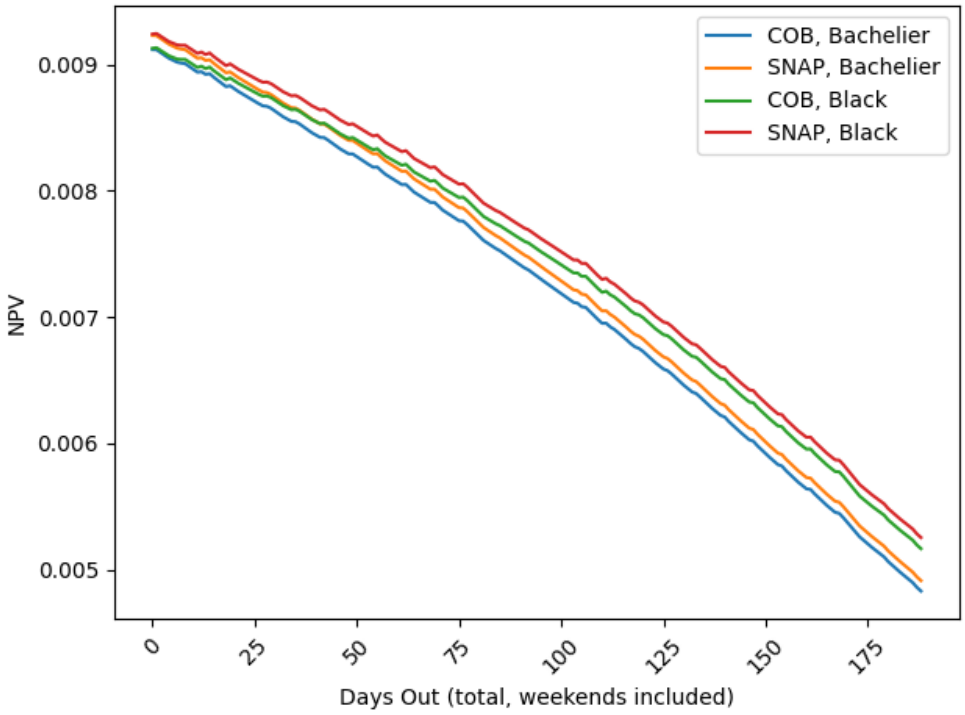


Figure 4.2: Curve/model NPV at projected date, starting at 12 March 2026

This figure visualizes the decay of the projected price and the differences in projected pricing between combinations of model and data. We can see the varying time-decay behaviors of the Black and Bachelier models. They start off more dependent on data source, but end up more separated by model type, where the Black model values the swaption higher at lesser times to maturity.

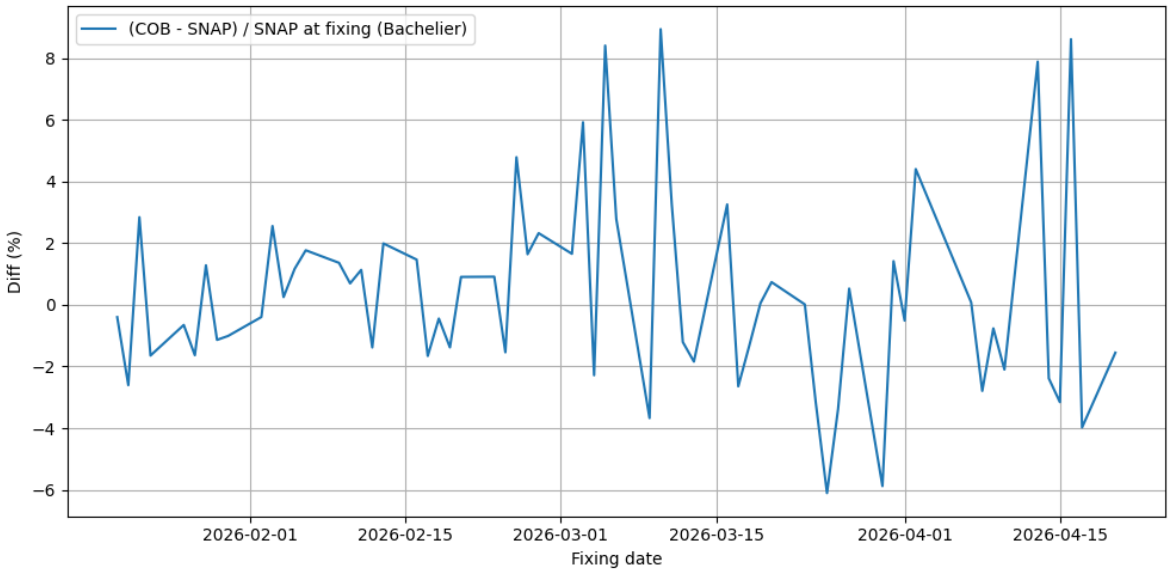


Figure 4.3: COB vs SNAP at fixing date

This figure visualizes the percentual difference between the COB-data and the SNAP-data using the Bachelier model. The difference is calculated as $\frac{COB-SNAP}{SNAP}$ and each datapoint is calculated at fixing with no projections. On visual inspection, there seem to be no trends, but there do seem to be occasional large spikes. The largest is about 8.5%.

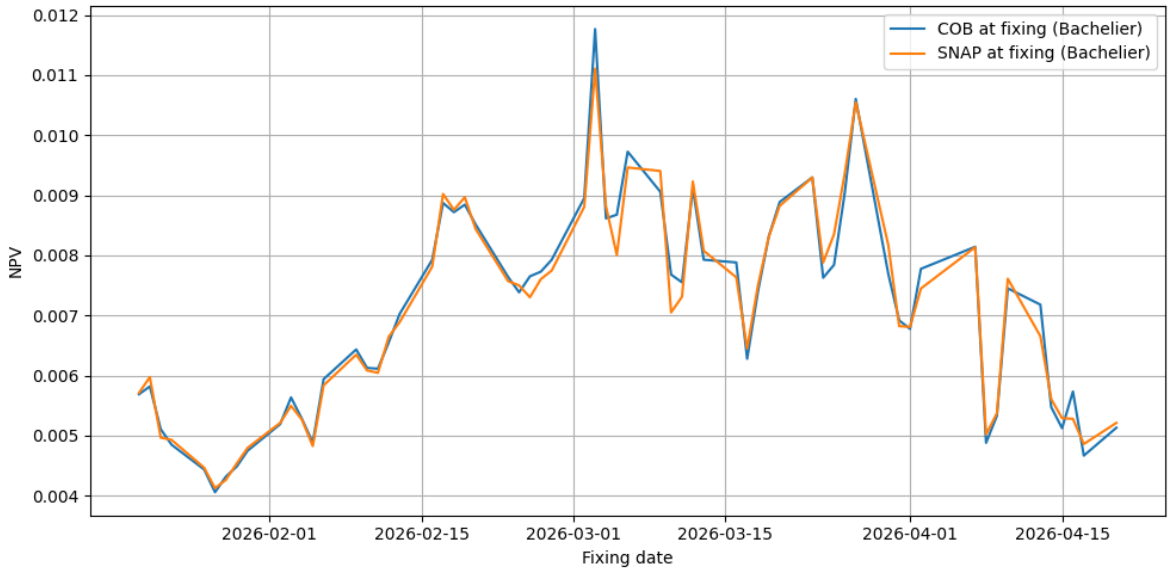


Figure 4.4: COB, SNAP NPV at fixing

This figure shows the NPV-plots for COB-data and SNAP-data using Bachelier, overlaid. The occasional bigger intraday differences are here seen in context of the abolute prices.

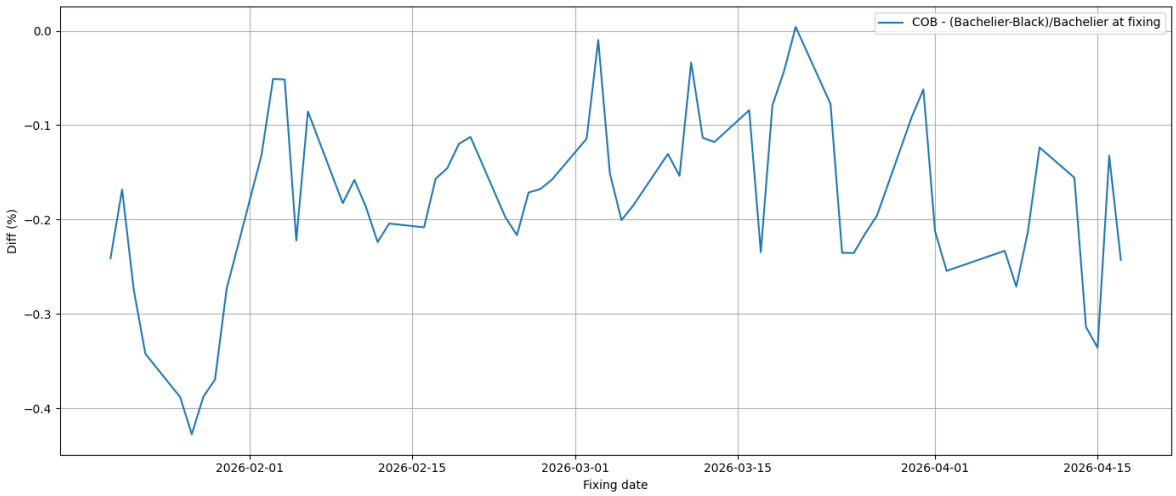
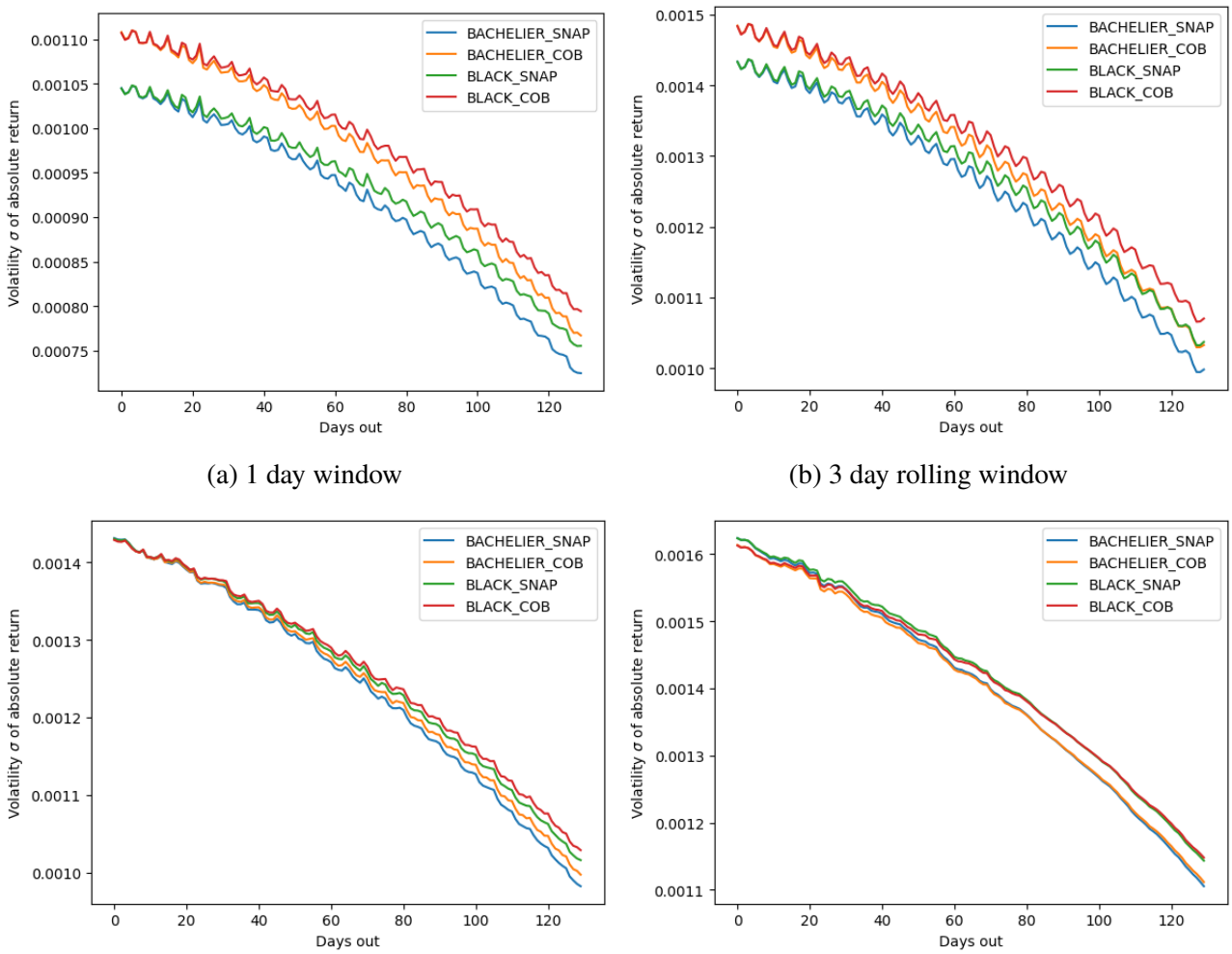


Figure 4.5: Bachelier/Black difference at fixing date

This figure visualizes the percentual difference between the Black model and the Bachelier model using COB-data. The difference is calculated as $\frac{Bachelier-Black}{Bachelier}$ and each datapoint is calculated at fixing with no projections. The differences are quite limited at fixing, with Black seeming to systematically price the swaption higher.



(a) 1 day window

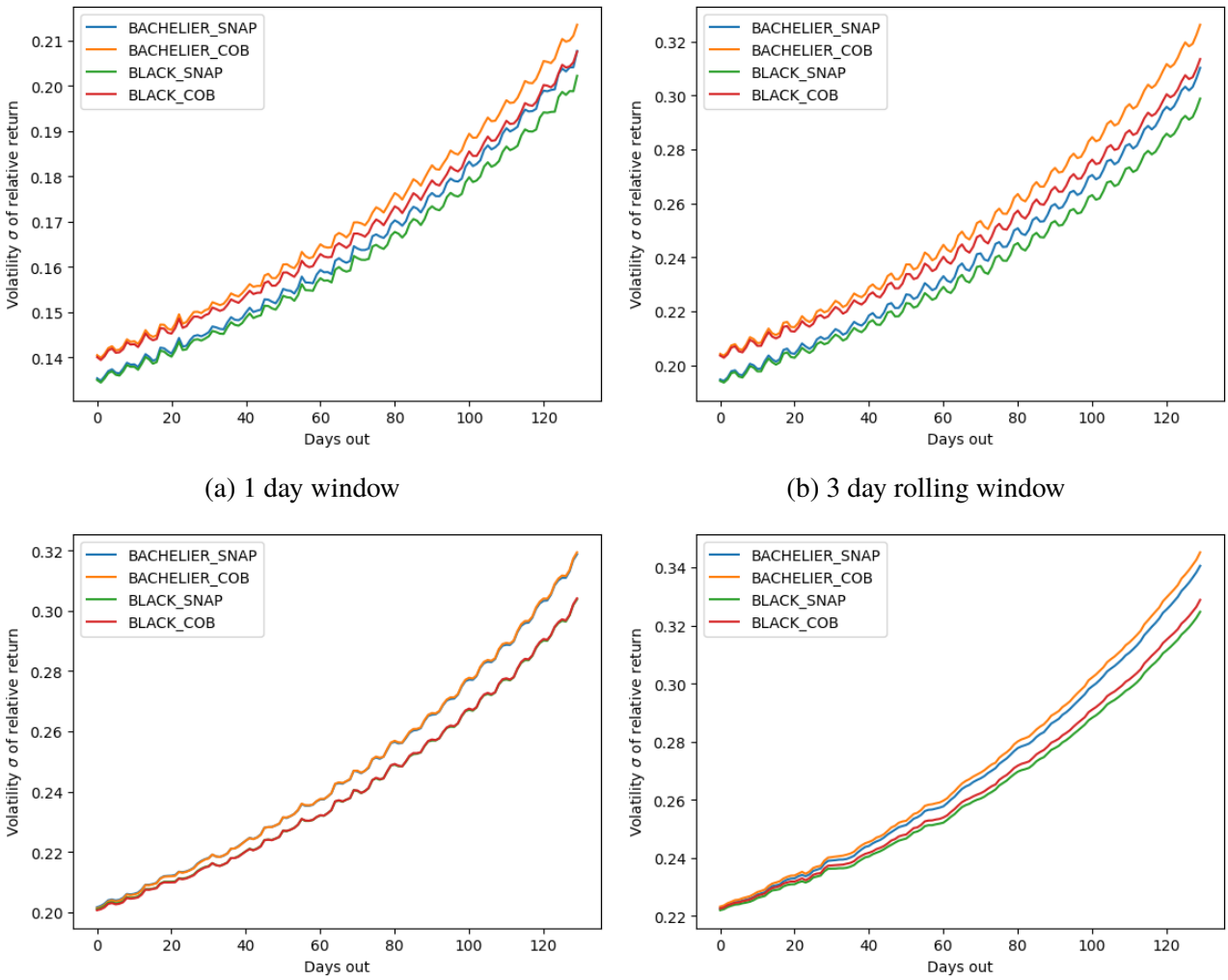
(b) 3 day rolling window

(c) 4 day rolling window

(d) 5 day rolling window

Figure 4.6: Rolling absolute volatility against price projection length (Days out). Volatility calculated against the largest possible span of fixing dates in the dataset, 63 days.

This figure visualizes the rolling absolute volatilites calculated on the projected time series. There seems to be some regular "jaggedness" in the first three subfigures. This dissapears at the 5-day window plot. The SNAP/COB-agreement at the 4-day window seems to be a random artifact.



(a) 1 day window

(b) 3 day rolling window

(c) 4 day rolling window

(d) 5 day rolling window

Figure 4.7: Rolling relative volatility against price projection length (Days out). Same smoothness at window = 5. Volatility calculated against the largest possible span of fixing dates in the dataset, 63 days.

This figure visualizes the rolling relative volatilites calculated on the projected time series. The jaggedness still exists, implying it originates from the weekly absolute returns. It dissapears at window size 5 again. The relative volatilities are increasing because of the swaption being out of the money and so lower in price the closer to expiry it is (more days out). From the graphs we can draw the conclusion that this effect is stronger than the decreasing effect of the absolute differences.

Category	Original	Convention	Mean Δ (%)
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30/360 (Bond Basis) - Bachelier	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30/360 (Bond Basis) - Black	0.1819
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (Eurobond Basis) - Bachelier	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (Eurobond Basis) - Black	0.1819
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (ISDA) - Bachelier	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (ISDA) - Black	0.1819
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/360 - Bachelier	13.9821
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/360 - Black	12.3249
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/365 (Fixed) - Bachelier	0.6000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/365 (Fixed) - Black	0.7038
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30/360 (Bond Basis) - Bachelier	0.6914
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30/360 (Bond Basis) - Black	0.7718
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (Eurobond Basis) - Bachelier	0.6914
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (Eurobond Basis) - Black	0.7718
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (ISDA) - Bachelier	0.6915
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (ISDA) - Black	0.7719
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/360 - Bachelier	−10.9618
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/360 - Black	−9.1915
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/365 (Fixed) - Bachelier	0.0000
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/365 (Fixed) - Black	0.1819
Stibor compounding	Compounded - Bachelier	Compounded - Bachelier	0.0000
Stibor compounding	Compounded - Bachelier	Compounded - Black	0.1819
Stibor compounding	Compounded - Bachelier	CompoundedThenSimple - Bachelier	148.1536
Stibor compounding	Compounded - Bachelier	CompoundedThenSimple - Black	127.8458
Stibor compounding	Compounded - Bachelier	Continuous - Bachelier	−11.6311
Stibor compounding	Compounded - Bachelier	Continuous - Black	−9.7574
Stibor compounding	Compounded - Bachelier	Simple - Bachelier	148.2490
Stibor compounding	Compounded - Bachelier	Simple - Black	127.9292
Stibor compounding	Compounded - Bachelier	SimpleThenCompounded - Bachelier	0.0491
Stibor compounding	Compounded - Bachelier	SimpleThenCompounded - Black	0.2234
Swaption BDC	ModifiedFollowing - Bachelier	Following - Bachelier	0.0000
Swaption BDC	ModifiedFollowing - Bachelier	Following - Black	0.1819
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedFollowing - Bachelier	0.0000
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedFollowing - Black	0.1819
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedPreceding - Bachelier	0.0036
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedPreceding - Black	0.1850
Swaption BDC	ModifiedFollowing - Bachelier	Preceding - Bachelier	0.0036
Swaption BDC	ModifiedFollowing - Bachelier	Preceding - Black	0.1850

Figure 4.8: Table of the mean percentage differences between the varied conventions and the original conventions. In the swaption the fixed leg convention was varied.
DCC = Day Counter Convention
CC = Compounding Convention
BDC = Business Day Convention

This table contains mean percentage differences caused by varying conventions, along with the specification of how they were varied (Original convention, category). It can be seen that the greatest differences were caused by Actual/360, CompoundedThenSimple, Simple and Continuous.



(a) Significant differences - Fixed Leg Daycount Convention

(b) Significant differences - Stibor Daycount Convention

(c) Significant differences, Large - Stibor Compounding Convention

(d) Significant differences, Very Large - Stibor Compounding Convention

Figure 4.9: Significant relative NPV differences by convention at fixing day.

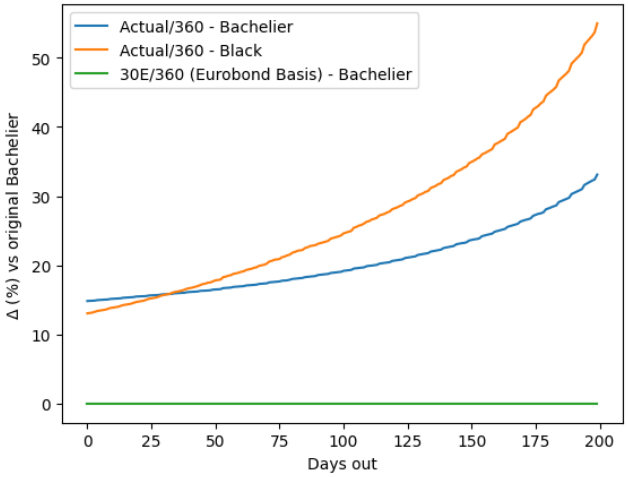
In this figure the NPVs of the conventions causing the large relative changes were plotted against their original (the baseline) at fixing date. We can see that the pricing dynamics seem to mimic the NPV-evolution at fixing date.

Category	Original	Convention	Mean Δ (%)	Δ at 199 d.o. (%)
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30/360 (Bond Basis) - Bachelier	0.0000	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30/360 (Bond Basis) - Black	8.3344	21.9824
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (Eurobond Basis) - Bachelier	0.0000	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (Eurobond Basis) - Black	8.3344	21.9824
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (ISDA) - Bachelier	0.0000	0.0000
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	30E/360 (ISDA) - Black	8.3344	21.9824
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/360 - Bachelier	20.5059	33.1056
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/360 - Black	27.1836	54.9599
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/365 (Fixed) - Bachelier	0.8527	1.3212
Swaption DCC	30E/360 (Eurobond Basis) - Bachelier	Actual/365 (Fixed) - Black	9.1232	23.3183
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30/360 (Bond Basis) - Bachelier	0.9704	1.5479
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30/360 (Bond Basis) - Black	9.2194	23.5306
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (Eurobond Basis) - Bachelier	0.9704	1.5479
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (Eurobond Basis) - Black	9.2194	23.5306
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (ISDA) - Bachelier	0.9704	1.5479
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	30E/360 (ISDA) - Black	9.2194	23.5306
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/360 - Bachelier	−15.7513	−24.0472
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/360 - Black	−6.1307	−2.4704
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/365 (Fixed) - Bachelier	0.0000	0.0000
Stibor curve DCC	Actual/365 (Fixed) - Bachelier	Actual/365 (Fixed) - Black	8.3344	21.9824
Stibor compounding	Compounded - Bachelier	Compounded - Bachelier	0.0000	0.0000
Stibor compounding	Compounded - Bachelier	Compounded - Black	8.3344	21.9824
Stibor compounding	Compounded - Bachelier	CompoundedThenSimple - Bachelier	290.7755	656.9146
Stibor compounding	Compounded - Bachelier	CompoundedThenSimple - Black	267.3918	628.0308
Stibor compounding	Compounded - Bachelier	Continuous - Bachelier	−16.8957	−25.7109
Stibor compounding	Compounded - Bachelier	Continuous - Black	−7.1901	−4.1952
Stibor compounding	Compounded - Bachelier	Simple - Bachelier	290.8593	657.1384
Stibor compounding	Compounded - Bachelier	Simple - Black	267.4671	628.2357
Stibor compounding	Compounded - Bachelier	SimpleThenCompounded - Bachelier	0.0279	0.0445
Stibor compounding	Compounded - Bachelier	SimpleThenCompounded - Black	8.3599	22.0270
Swaption BDC	ModifiedFollowing - Bachelier	Following - Bachelier	0.0000	0.0000
Swaption BDC	ModifiedFollowing - Bachelier	Following - Black	8.3344	21.9824
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedFollowing - Bachelier	0.0000	0.0000
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedFollowing - Black	8.3344	21.9824
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedPreceding - Bachelier	0.0052	0.0081
Swaption BDC	ModifiedFollowing - Bachelier	ModifiedPreceding - Black	8.3392	21.9906
Swaption BDC	ModifiedFollowing - Bachelier	Preceding - Bachelier	0.0052	0.0081
Swaption BDC	ModifiedFollowing - Bachelier	Preceding - Black	8.3392	21.9906

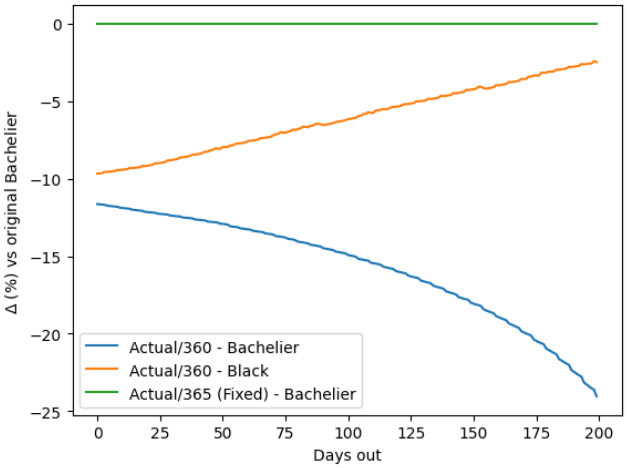
Figure 4.10: Table of the mean and final percentage differences between the varied conventions and the original conventions when projecting the price 200 days out from 20 Jan 2026. In the swaption the fixed leg convention was varied.
DCC = Day Counter Convention
CC = Compounding Convention
BDC = Business Day Convention
d.o. = Days Out

This table displays the effects of conventions on NPV-projections. It displays the percentage differences over the days out, along with the specification of how they were varied (Original convention, category), as for the previous table. It also

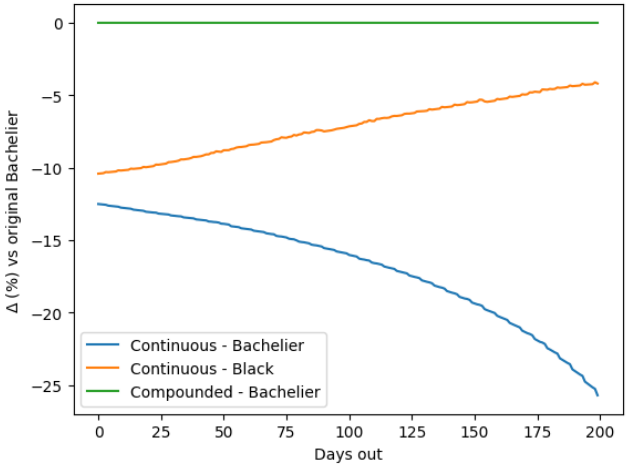
displays the percentual difference at the maximum days out. It can be seen that the greatest differences were caused by Actual/360, CompoundedThenSimple, Simple and Continuous, along with all Black models.



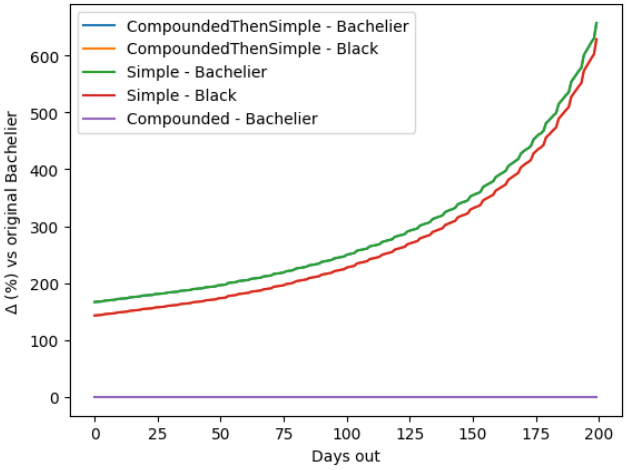
(a) Significant differences - Fixed Leg Daycount Convention



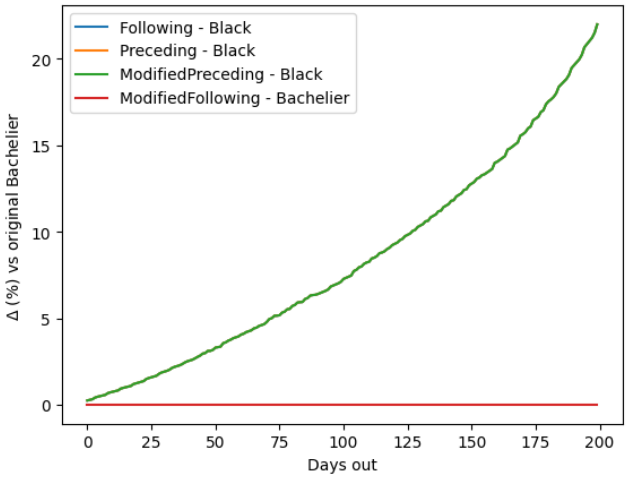
(b) Significant differences - Stibor Daycount Convention



(c) Significant differences, Large - Stibor Compounding Convention



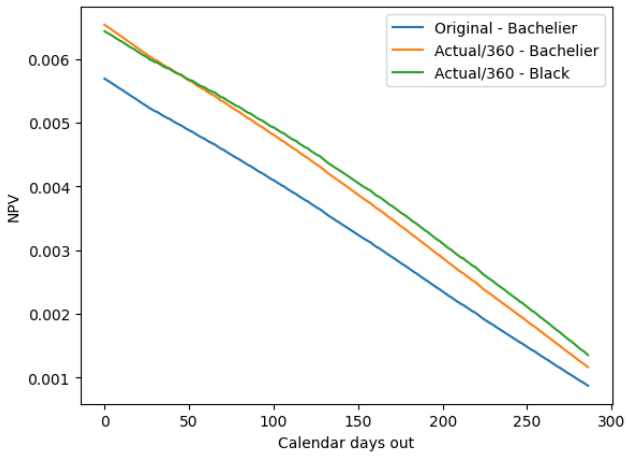
(d) Significant differences, Very Large - Stibor Compounding Convention



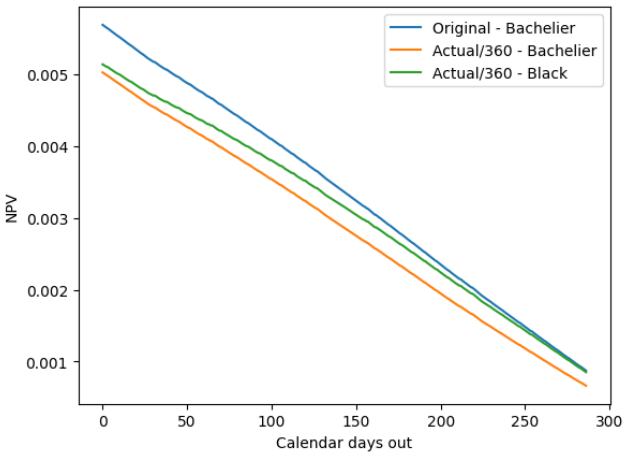
(e) Significant differences - Stibor Business Day Convention

Figure 4.11: Relative NPV values by convention as projections from 20 Jan 2026.

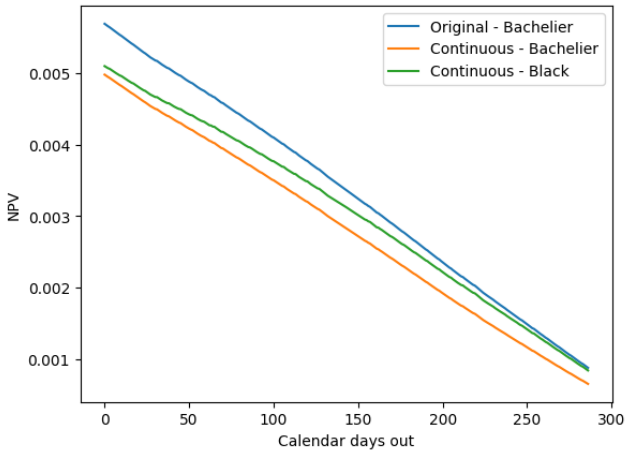
In this figure the relative NPVs of the conventions causing the large relative changes in the last table were plotted against their original (the baseline) at days out. We can see some interesting pricing dynamics appear. In some plots the Black/Bachelier models move in the same direction, while in others they diverge. For the most part, the Black model values the swaption higher than Bachelier, but sometimes it doesn't (subplot (d)).



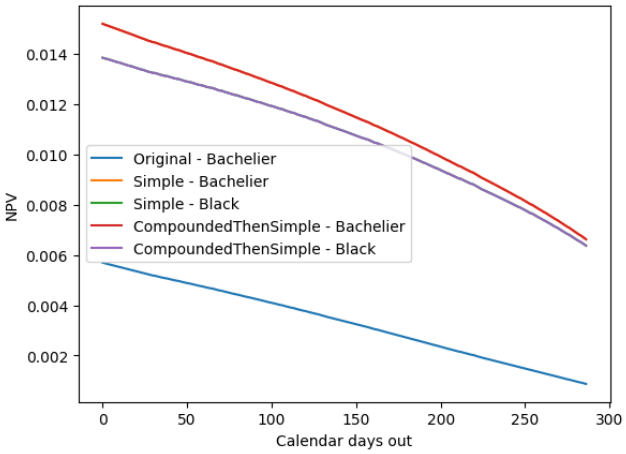
(a) Significant differences, full NPV - Fixed Leg Daycount Convention



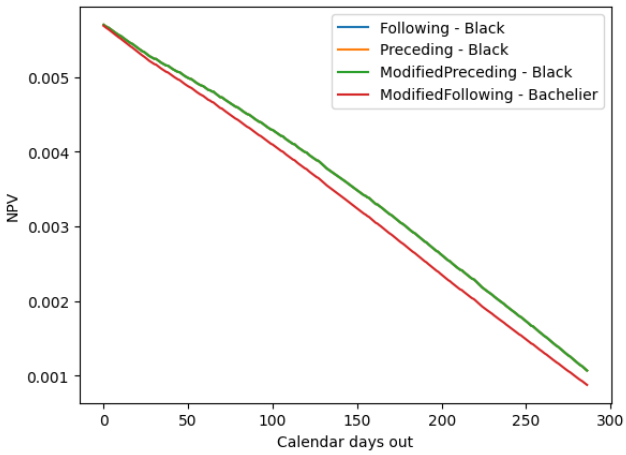
(b) Significant differences, full NPV - Stibor Daycount Convention



(c) Significant differences, full NPV, Large - Stibor Compounding Convention



(d) Significant differences, full NPV, Very Large - Stibor Compounding Convention



(e) Significant differences, full NPV - Stibor Business Day Convention

Figure 4.12: Absolute NPV values by convention as projections from 20 Jan 2026.

This plot shows the absolute NPV for the projected price for the conventions causing the large relative changes at calendar days out. The behavior of the absolute NPVs seem to decay with time as expected.

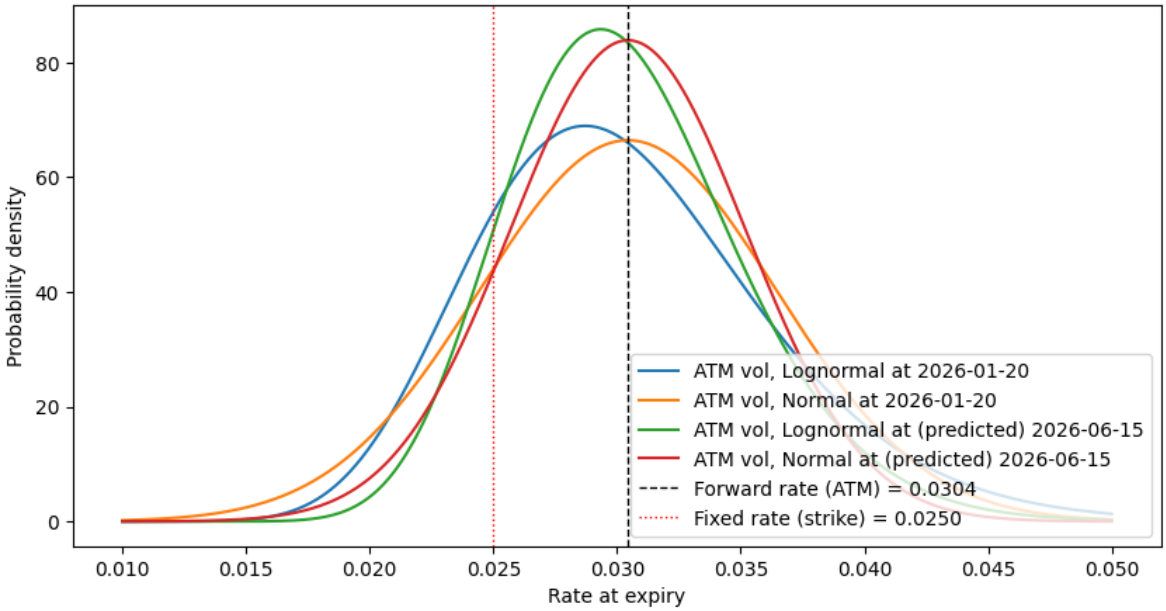


Figure 4.13: The lognormal and normal probability density functions as implied by ATM pricing volatility

This figure shows the distributions of the probability density functions of the underlying swap for Black and Bachelier-models at ATM volatility. We can see how the Black model has the probability-mass shifted to the left, towards lower rates. We can also see the symmetric distribution of the Bachelier model. We can also see that the reduced time to maturity reduces the volatility, concentrating the distributions in addition to skewing the lognormal distribution. The decay in price (Theta) can be visualized as the probability mass (area under the graph) to the left of the strike rate shrinking, along with the graph shifting rightwards (payoff shifting lower) for the projected distributions.

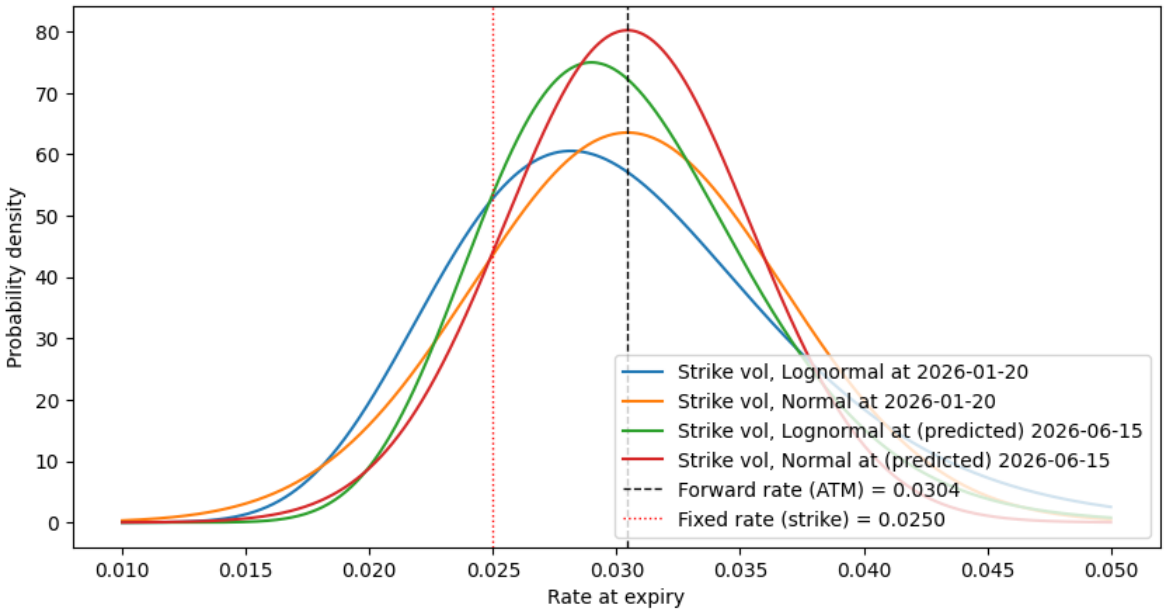
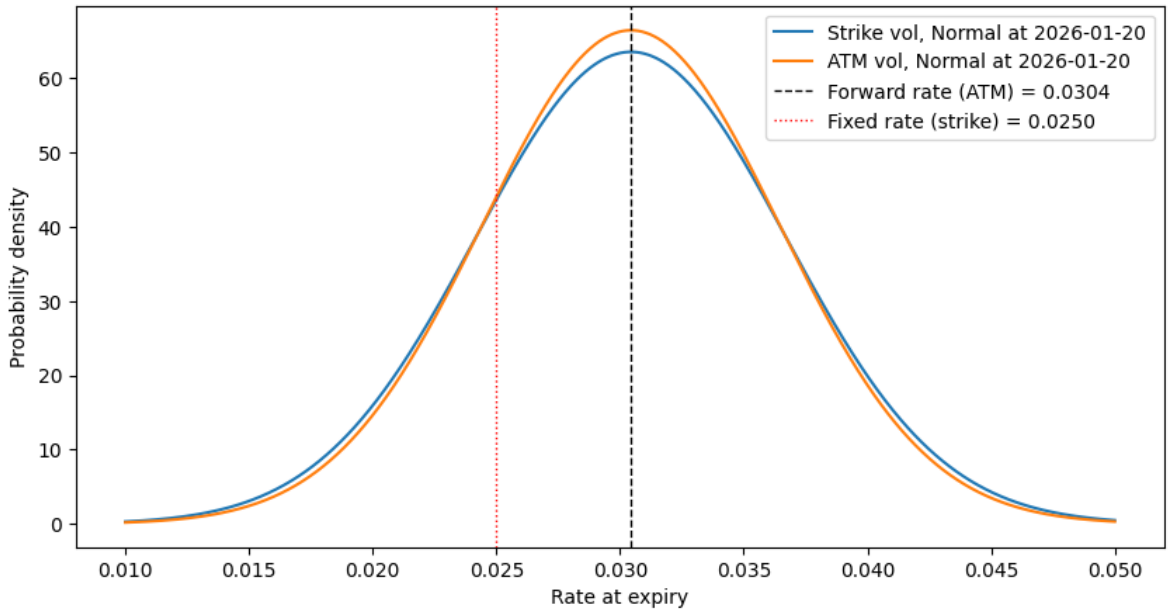
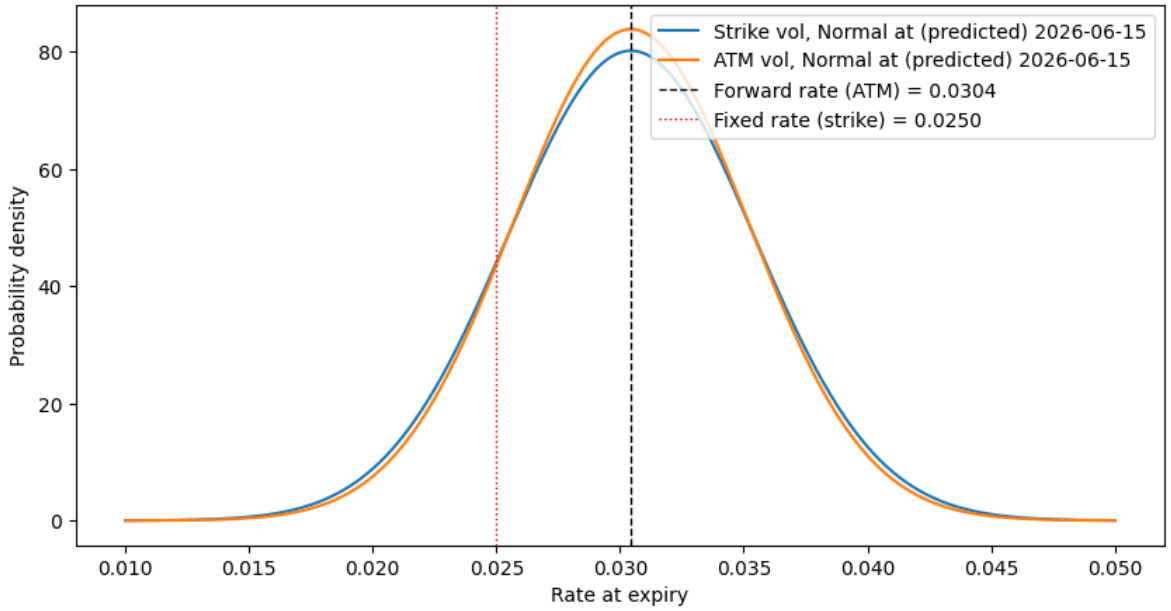


Figure 4.14: The lognormal and normal probability density functions as implied by actual shift (out of the money) vol

This figure shows the distributions of the probability density functions of the underlying swap for Black and Bachelier-models at actual out-of-the money volatility (per 2026-01-20). Using the Fixed/Forward rate-lines as a visual aid, we can see that the distribution for the Bachelier model has slightly widened, while the Black model has widened and also gained in skew. The following figures will show this effect in more clarity. We can also see the same time-to-maturity (TTM)-shift behavior as in the previous figure, although the correspondence with the earlier NPV-graphs is not clear from the plots. The estimated relative shifts in NPV from these graphs will be presented in the final table of this section.



(a) The normal probability density functions as implied by ATM and Strike pricing volatility at fixing date



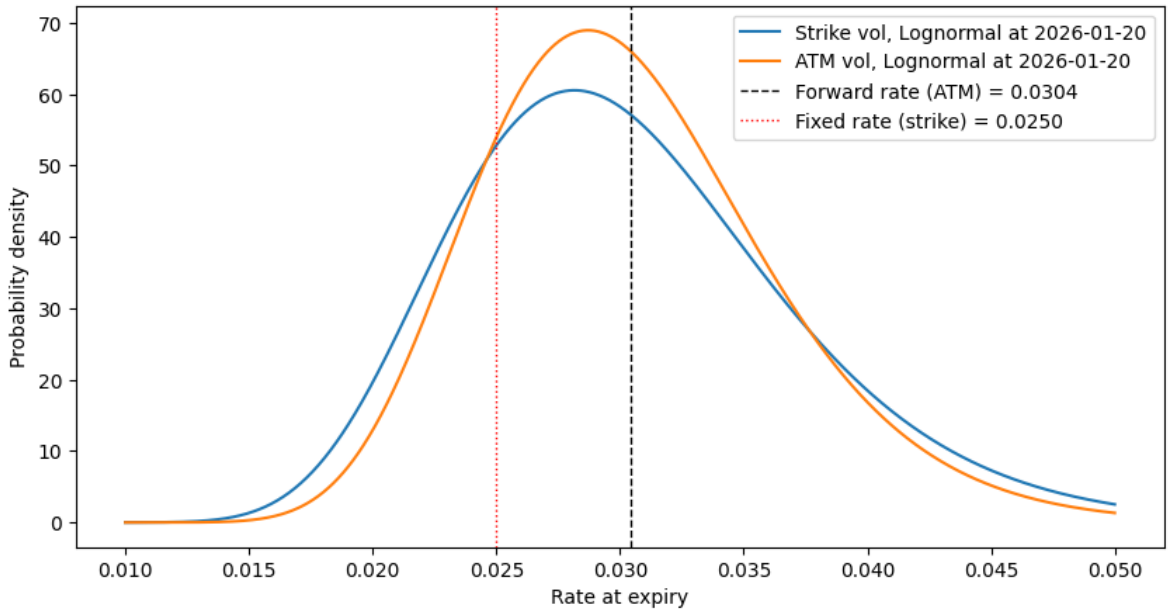
(b) The normal probability density functions as implied by ATM and Strike pricing volatility at a projected date

Figure 4.15: Comparison of normal distributions by volatility

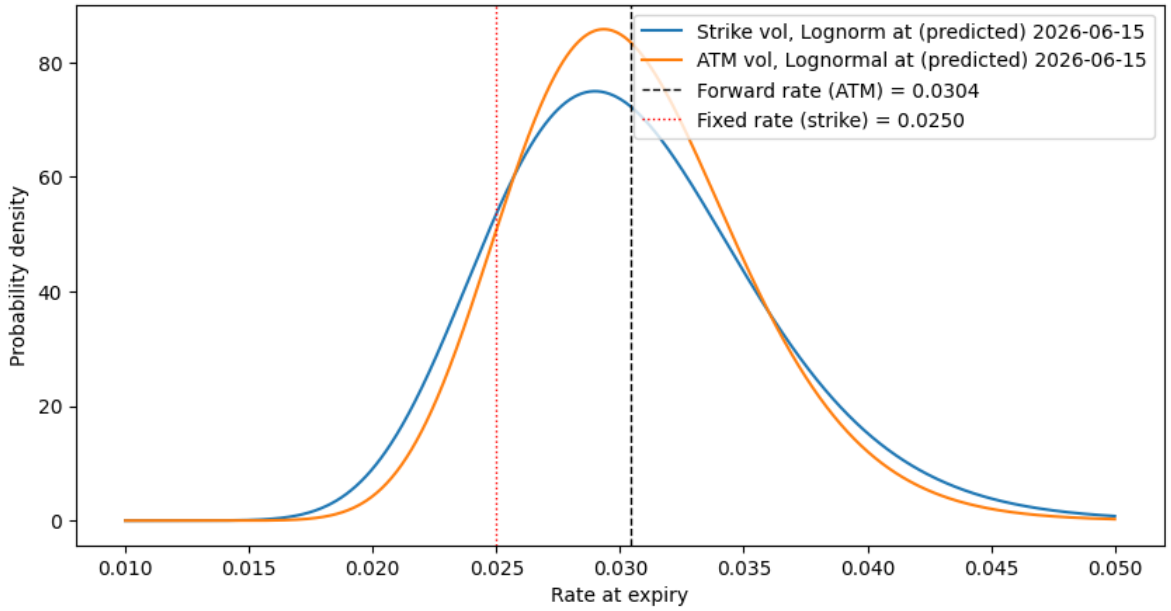
This figure contains close-up comparisons of the cross-plot (Strike vs ATM) intra-

model differences in figures 4.13 and 4.14 for normal (Bachelier) distributions, for both fixing date and the "predicted"/projected distributions.

Due to the "smile"-effect, the volatilities increase when moving away from the center (ATM price/strike), and we can see the effect of our larger actual strike implied volatility by the widening distributions. As discussed in section 2.1.12, in theory (model assumption) the volatilites do not depend on the strikes of the options, but in practice non-ATM options generally have greater implied volatilites.



(a) The lognormal probability density functions as implied by ATM and Strike pricing volatility at fixing date



(b) The lognormal probability density functions as implied by ATM and Strike pricing volatility at a projected date

Figure 4.16: Comparison of lognormal distributions by volatility

This figure contains close-up comparisons of the cross-plot (Strike vs ATM) intra-

model differences in figures 4.13 and 4.14 for lognormal (Black-76) distributions, for both fixing date and the "predicted"/projected distributions.

In addition to a reduction in "width", we can also see an increased skew as a consequence of the greater implied volatility at strike. This follows naturally from the formulation of the lognormal distribution for the Black-76 model, where $\mu = ln(s) - \frac{\sigma^2}{2}(T-t)$.

Model	**Date Start**	**Date Prediction**	$\Delta \int_{-0.01}^{Strike\ Rate} ITM\ Rate(x) \cdot f_{model\ density}(x) \cdot dx$
Bachelier (Normal)	2026-01-20	2026-06-15	$\approx -4.8 \cdot 10^{-4}$
Black-76 (Lognormal)	2026-01-20	2026-06-15	$\approx -3.2 \cdot 10^{-4}$

Figure 4.17: Table of the models and the *ITM Rate*-weighted probability-shift as a function of a TTM-shift. The integral is numerically approximated using rectangles and 1000 steps on the range $[-0.01, 0.0250]$, where the first value is chosen so that the left tail is more or less completely flat and the last value being the strike rate.

x = Rate at expiry

$f_{model\ density}(x)$ = Density function of the model

$ITM\ Rate = Strike\ rate - x$

In this table, the values calculated should correspond to the loss of payoff as a consequence of the TTM-shift. The calculations scale the ITM-rate (rate amount difference between the fixed and floating legs we earn as the purchaser of a receiver swap) by the probability of that floating rate at expiry. As we can see, the Bachelier model shifts more downward to a greater degree compared to the Black-76 model, which corresponds to our earlier NPV-graphs.

Chapter 5

Analysis and Discussion

An important reminder is that the current swaption examined is “out of the money”. This is central to the results in the following section.

The first figure in the results, figure 4.1, visualizes the time decay of the fixing date pricing. The visualized behavior is reasonable as we are using the formulas to project using our current data by reducing the time to maturity, which results in a lower probability of reaching the strike price (shorter time). It was also commented on in section 3.2.2.

5.1 Theoretical Analysis of Results

5.1.1 COB/SNAP - NPV

One could assume that using data from 2 hours ago to price an instrument should not have that great of an effect. While the differences in figure 4.4 seem small, looking at figure 4.3 the effect of an intra-day difference in the time of data sampling seems to occasionally produce very large NPV-differences. Generally, the 2-hour intraday difference in the floating curve is in fact not actually that large, absolutely speaking. However, since it creates a small increase (or decrease) in the spread between the fixed and floating legs of the swap, and the swap has a tenor of 10 Years, this large NPV difference could be explained by this small absolute rate difference having a very long time to exponentially accumulate. Furthermore, swaptions on underlying swaps with maturities of 30 years are not at all uncommon, and presumably the results would be even more exaggerated, however that remains to be confirmed.

An important caveat is that here only the floating curve has SNAP data, so any co-variance with the discounting curve is not visible. Still,

instinctively one might not think that a 2 hour difference in time of data sampling should make any difference at all, and if a floating curve sampled at a different time is all that is easily available, it is not impossible to imagine pricing being completed with this possibly problematic result. Generally, this could be extended to be seen as an illustrative example of how sporadic errors/oversights caused by undefined routines could result in quite substantial over/under exposure, or too much/too little posted/requested margin/collateral.

5.1.2 Black-76/Bachelier - NPV

The model comparison seems to be somewhat multifaceted, as hinted by figures 4.11 and 4.12. Beginning with figure 4.2, it is a plot of the projected NPVs from the initial date 12 March 2026. We can see that this initial date results in a relatively large initial NPV-difference caused by COB/SNAP. Initially the Black and Bachelier prices are quite close, but as we project out they seem to diverge by model, with the Black model pricing the instrument higher than Bachelier. The first explanation for this would be that the two models have different time-to-maturity-dependent behaviors. This can be seen upon visual inspection of the formulas in sections 2.2.7 and 2.2.8. However, we can also consider the probability distributions in order to understand this difference. The swaption is an out of the money receiver, meaning the floating rate exceeds the fixed rate. In order to value the swaption higher, the model would need a distribution which implies a higher probability of lower rates, weighted by payoff. Looking at the distributions in figures 4.13 and 4.14 we can see that this is precisely the case with the lognormal (Black) distribution. The skew points towards lower rates, meaning we have a larger probability mass to the left of the strike compared to the normal (Bachelier) distribution which has the center of probability mass coinciding with the mean value (forward rate). However, the prices are equivalent at a time-to-maturity (TTM)-shift of 0 (fixing date), which is explained by the one-price assumption in deriving the implied lognormal volatility for the Black model. As such these model differences are “normalized” at fixing and possibly misleading; comparing the relative distribution movements (weighted by payoff) could be more relevant for actually explaining the difference in NPV when projecting using a pure TTM-reduction. This relative shift has been tabulated in figure (table) 4.17, where we can indeed see that the payoff weighted probability decreases more for the Bachelier model, which agrees with our TTM-shift NPV plots where the Bachelier model seems to decay more with a reduction in TTM than the Black-76 model does.

Interestingly, looking at figure 4.5, we can see that this seems to hold at fixing date (no projection) too, albeit at a much smaller scale. This result is somewhat curious as the volatility for the Black model (and as such the pricing) has been derived using the one-price assumption, but it could be explained by the effect of differences in *non-nonlinearities* between the models co-interacting with the *linear* interpolation of the volatility cubes.

5.1.3 Volatility

In figures 4.6 and 4.7 absolute and relative volatility have been calculated for each possible prediction offset (days out, reducing TTM). In other words, return time-series are created for each fixed days out offset, while varying the fixing date, and volatility being calculated over these time series. These are then plotted for all model/data combinations and for various rolling window sizes. The difference in incline between the relative and absolute returns is caused by the fact that the NPV of the out of money swaption is decaying with decreasing time to maturity (increased days out) because the swaption is out of the money. This decay in NPV with increasing days out is then seemingly greater than the decrease in absolute returns. This also aligns with the idea of projected prices getting relatively more sensitive as shifted time to maturity nears 0, and while the absolute vol drops due to the sharply dropping NPV, the relative volatilites increase due to the movements becoming more extreme in the relative sense.

One interesting aspect worth noticing is the “jaggedness” more or less dissapearing at the 5-day rolling window. It seems that there is still some minor convention not fully correctly configured which is only relevant intra-week. Another interesting thing is how the COB/SNAP derived volatilites seem to coincide at the 4-day rolling window. This could be some missed convention once again, but this effect was not reproducible when testing other window sizes. As such it seems to be some type of coincidence/artifact.

The relative overpricing of the Black model compared to the Bachelier model when purely shifting the TTM lower is also visible in the absolute volatilites. The absolute Black model volatilites grow larger (relatively). This is not unexpected considering the NPVs for the Black models grow relative to the Bachelier model NPVs when performing the same shift. This pattern inverts when plotting relative volatilites. This could have to do with the NPV of the Black models (the denominator) being greater in some certain proportion.

5.1.4 Convention Differences at Fixing Dates

Figure (Table) 4.8 contains all varied conventions (along with the original), together with the mean percentage difference over all fixing days. We can see that the most exaggerated differences are caused by the compounding conventions connected to the "Simple" formula. Both the Simple and CompoundedThenSimple are more or less purely "Simple" (with the latter including one period of "Compounded"). Since these are applied on the floating rate, it seems like it drastically undervalues the total interest paid by the floating leg. This is reasonable if we simply observe the formulas or plot them in a graphing calculator. The "Compounded" convention has higher values when the time t increases.

The second large difference is caused by the Act/360 Day Count Convention. In the floating curve it is compared with Act/365 which has a greater denominator. This implies that the floating curve coupons/rate will be greater, which in turn implies that our receiver swaption will be further out of the money. We can also see that the effect when acting on the fixed leg of the swaption is more or less inverted, the fixed coupons (which we receive) are calculated using a smaller denominator resulting in greater fixed coupons. This table also implies the other Day Count Conventions result in somewhat similar day count fractions, while Act/360 is an outlier.

The largest of these differences have been plotted in figure 4.9. We can see that all the resulting difference plots seem to be images, or mirror images, of the absolute NPV at fixing plot in figure 4.4. This could be caused by one of the legs generating the variance (the floating leg), and by the convention in these experiments only affecting one leg at a time. As such, we get an amplification in the proportional pattern attributed to one of the legs, relative to the original model.

Interestingly, the Black model seems to value the swaption lower compared to the Bachelier when the the difference is positive, but higher when the difference is negative (i.e. a lower absolute percentage NPV difference). This could possibly partially be explained by the percentage difference being calculated against the NPV of the original swaption valued using Bachelier, but the exact dynamics of this relation could warrant further investigation. It seems, however, that the Bachelier model responds more strongly to changes in convention, at least in the examples plotted.

Finally, one can consider the "sporadic" appearance of extreme mispricings (sporadic in the sense of randomly picking conventions to examine, or conventions "randomly" getting misconfigured). As exemplified by the

configuration of the current swaption in use in section 3.1, there are very large amounts of different conventions in different sub-instruments. At the same time, it can be difficult to fully automate the computerization of custom OTC-derivatives due to their bespoke nature. These two conflicting factors could be a large challenge when designing both efficient and safe Mark-to-Model-pricing-processes.

5.1.5 Convention Differences at Predicted Dates

Figure (table) 4.10 contains the analogous table to above, but for projections instead. Here the values are the mean difference over the extending projection, and the difference at the end of the projection. Here we can see the same conventions as previously, but with far exaggerated differences, especially at the maximum TTM-shift/days out. We can also see that more or less all Black models have differences compared to the original Bachelier models when shifting TTM. Inspecting figures 4.11 we see some puzzling results, with Black/Bachelier diverging or crossing for example. It seems like the Bachelier model exaggerates the convention differences more at fixing day (as in the previous section, 5.1.4, with the results for fixing days), resulting in larger absolute percentage differences at 0 days out, but when increasing the TTM-shift behavior becomes harder to interpret. It could possibly be caused by the relative behavior when the TTM is nearing 0, or when the is NPV nearing 0. Looking at figure 4.12 it seems that this indeed is the case, with the NPVs decaying and Black model resulting in a higher NPV in most plots. Black-76 valuing the swaption higher was also the case for the standard swaption, so this once again agrees with the relative decreases tabulated in table 4.17, where the Black model loses less probability mass ITM (in the money) weighted by payoff. Still, the intermediate dynamics seem somewhat sporadic and incomprehensible and could warrant further investigation.

Another aspect to note is that graph (d) does not seem to agree with the above discussion, as the Bachelier model is returning greater NPVs than the Black-76 model. This could be due to the differences in nonlinearities between the models as mentioned in section 5.1.2 (the models using either a lognormal volatility cube or a normal volatility cube, but both being linearly interpolated). It could also have to do with more complicated strike/TTM/option nonlinearities co-interacting with the models/distributions, such as something like the difference between the normal distribution and the skewed (lognormal distribution (illustrated in figures 4.13, 4.14, 4.15 and 4.16)) resulting in non-linear effects when internal aspects of the swaption

(such as the returns) vary. Indeed, these non-linearities could possibly help explain the previously mentioned intermediate dynamics too. However, these are just ideas/conjectures and plot (d) warrants further investigation, along with the above-mentioned intermediate dynamics.

5.1.6 Comments on Mark-to-Model

The Mark-to-Model procedure is based on various market/price-dynamic assumptions and the general theoretical construction, which are not necessarily always tested/confirmed. As such it can be difficult to say that these are "real" prices in a Mark-to-Market sense. Furthermore, understanding all complications, assumptions and caveats seems extremely important when Marking-to-Model.

One example of such a caveat/assumption is the TTM-reduction-projection used in this analysis. It assumes all other inputs are fixed which might not be the expected market behavior. The expected values and behaviors of the volatilites (for example) could be calculated mathematically, but this once again depends on what assumptions we make regarding the market behavior and what estimations we are ready to make in our calculations.

Another aspect worth noting is the contrast between the formulas assuming constant volatility and the actual market implied volatilites differing by strike, time to maturity, etc. As noted by Hagan et al. (in the context of the Black model), "Changing the volatility σ_B means that a different model is being used for the underlying asset for each K and t_{ex}." [17]. We can see an example of this effect in the plotting of the distributions gained from ATM vs Strike volatilites in figures 4.15 and 4.16, with the shifting distributions (by the increased volatility at Strike) illustrating another limitation of marking-to-model.

In the end, the ways the models can be correctly utilized, and the validity/possible use-cases of the results are strictly defined by the assumptions made during model derivation and relevant theory. It seems that the involvement of theoretical expertise in the design of processes and in the possible ad-hoc usage of such Mark-to-Model tools could serve to greatly improve the risk resistance.

5.2 Applications to Industrial Economics and Management

From the above results it is clear that all three main points of examination (Data, Models, Conventions) can have large effects on the pricing of swaptions.

5.2.1 Data

The results illustrate the importance of using correctly snapped data, or rather the risks of forgoing doing so. While some might not consider a two-hour intraday difference in data sampling as something of consequence, the results have illustrated that swaptions can exaggerate this small difference and create large mispricings. In our case the COB/SNAP difference seemed somewhat centered around 0%, which may lead some to feel like it can be disregarded. Still, for actors with rare but large transactions, or if some trend-like behaviors start appearing intraday, mispricings could be substantial.

5.2.2 Models

The Black and Bachelier models agree closely at fixing date, but they start to differ increasingly with pure TTM-projections further out, which is likely due to the models containing varying assumptions concerning the underlying market dynamics. As such, extra vigilance in coordinating the bilateral valuation methods when entering into OTC-contracts would be prudent in cases where the prices are set regarding some future value.

It is also interesting to point out that while the occurrence of data issues, or convention issues, can be seen to be mostly rooted in human error and more classical organizational theory, the mispricing caused by future predictions using various models could be more innocuous/difficult to "catch" as they could be seen as being more closely connected to the requirement of possessing deeper related theoretical knowledge.

5.2.3 Conventions

Perhaps the most extreme errors with regards to the absolute mispricings amongst the ones examined in this thesis are the mispricings caused by incorrect conventions. The largest of these (worst case found in the results being an $\approx 650\%$ increase in predicted NPV) should reasonably not happen systematically as the mispricing should be found relatively early in the

lifecycle of the OTC-instrument. As such these very large errors should mainly concern actors entering into rare and large transactions where prices are needed before the bilateral communication (such as fund valuations, possible significant costs during large fund flows). With regards to systematic errors, assuming such mispricings are smaller in magnitude (as in our following theoretical example), they would reasonably have a larger risk of passing "under the radar", and as such could be more persistent and perhaps dangerous. This also goes for the other mispricings described in sections above. To illustrate; say an employee is unfamiliar with configuring instruments/data entry and somehow thinks the fixed swaption leg should have a day convention of Act/360. This is not a difficult error to make! It can be as little as the difference between a 0 and a 5 at the end of one of many other conventions when configuring an instrument, which could result in the systematic mispricing of the instrument. Or perhaps the internal software uses slightly different/unfamiliar convention names, which is not unusual as the ISDA guide itself names many aliases for the same conventions; old and new names for Act/365 being "Actual/Actual (ISDA)", "Act/Act (ISDA)", "Actual/365", "A/365", "Actual/Actual", "Act/Act", "Act/Act (ISDA)" [14] (not to be confused with "Act/Act (ISMA)", "A/360", "A/365F", etc.).

5.2.4 Costs

As previously mentioned, it seems reasonable to assume that very large differences in NPV have a larger chance of being found out long term, resulting in these being more connected to sporadic errors (and as such less exploitable costs). These could still be very dangerous/costly in the case of rare but large transactions where position valuation is be done before the bilateral/cleared "syncing" of prices. An example could be the valuation of funds; if a certain value is calculated and presented to customers it will be the basis of subscribtions, redemptions, etc., which will have to be corrected (possibly along with fines and renumeration) when the error is found out.

Regarding systematic errors, and connecting the example in section 5.2.3 to our results; if one systematically misconfigures the fixed leg of a swaption as Act/360, and assuming the collateral is proportional to the NPV, the counterparty to this transaction would be able to demand $\approx$ 10% more collateral (assuming them having gained the knowledge of this mispricing). This additional demanded collateral is interest bearing and as such it practically results in a free increase in the return of the instrument of $0.1 \cdot RFR$! On the other hand, if the fixed leg of the counterparty

was systematically misconfigured as Act/360 one could (if one were an unscrupulous actor) demand 10% more collateral, leading to the opposite situation. Since one enters the transaction assuming a NPV of 0, systematic mispricings can risk generating hidden costs associated with entering into such OTC-derivatives.

Of course, the actual process of margin/collateral can, and often is, more complicated than this, but the example illustrates an interesting point. Systematic undervaluing of own positions could for example, if known by a counterparty eager to earn some more interest in an unscrupulous manner, result in them systematically demanding more collateral, turning a fair deal ($NPV = 0$) into a losing deal at start, letting the other party enter into transactions without knowing that they are subject to systematically greater costs connected to said transactions.

For actors concerned with having hidden costs in the form of "unfairly" requested collateral a possible solution could be adding a screening of total collateral according to various instrument/other-dimensional breakdowns. It could be used to identify if one has been exposed to such hidden costs by identifying possible sources of statistically improbable/systematically large collateral requests. It should be mentioned that this recommendation was inspired by the final sentences in [5], where Melin discusses the interest in examining the unexpectedly large collateral outflows that were found.

Fianlly, connecting to Mark-to-Model more generally, the caveats and quirks connected to the assumptions and theory/construction result in mistakes/human error not necessarily mapping linearly to costs; mistakes that are easy to do can easily "blow up". As discussed in section 5.1.6, it follows that thorough theoretical knowledge and experience along with a thorough understanding of constructing processes which minimize the impact of human error (organizational theory) are imperative when designing and auditing Mark-to-Model processes.

Chapter 6

Conclusions and Future Studies

6.1 Conclusions

The purpose of this article was to identify possibly overseen behaviors and illustrate the ways they might cause mispricings. Both possible systematic and sporadic errors have been identified among these behaviors/routines in question. Their magnitudes have also been quantified for the specific example concerning our swaption. Hopefully these examples can serve as a reminder to always question established routines/shortcuts and seek constant improvement. The results can also serve as good examples for auditors or process designers to illustrate the danger of manual handling and innocuous-seeming routines/shortcuts. A small proposal was also presented regarding screening posted collateral amounts across various aspects.

The general takeaway of this thesis can be said to be that the risk of significant price-misalignments does in fact exist. Indeed, certain common measures related to more standard organizational theory and process design could likely be used to great effect. Outisde of general strict process design, however, the fact that the results seem to agree with the theory/models and could in hindsight likely have been derived directly from theory serves as an interesting example of another Industrial Economics and Management-conclusion made in this thesis; that the deep understanding of the relevant theoretical financial mathematics is very important, and perhaps paramount, not only for designing the processes to be failure proof but for the effective auditing of them. This holds even in the case of sporadic human mistakes, as the knowledge of the possibility of great mispricings should necessitate a

rigorous process to reduce the risk of such.

In summary, complicated Mark-to-Model setups seem to require a heightened deal of care, caution, theoretical knowledge and organizational expertise, with regards to process design, implementation and auditing.

6.2 Future Studies

Many aspects could be worth expanding upon in a future study. A greater amount of data could allow for more tail end events, perhaps giving a more clear and nuanced result. Expanding upon the amount of timestamps for snapped data or performing this analysis for other types of instruments could also prove fruitful. Further analyzing the exact dynamics of the intermediate model behavior in TTM-shifting could also be of interest.

Further iterating, consider the previously mentioned observation that the difficulty of mistakes doesn't generally map linearly to costs. As such, a more extensive search/analysis could help in identifying and specifying more types of risky behaviors. A more extensive study could lay the foundation for a compendium of observations relating to designing or auditing such processes, as auditors without financial-mathematical understanding might not be guaranteed to be able to perform an efficient audit relying solely on organizational theory and more general financial/pricing knowledge.

Chapter 7

Proprietary/Jointly Internally Developed Code, External Code Packages and AI Usage

The project utilizes proprietary code/knowledge connecting the data sources and their format with QuantLib usage. It has been developed at SEB Asset Management by not only me but also my co-worker (Coën Olofsson), with the occasional use of AI (in ways of routine enterprise code development, mostly debugging and obtaining knowledge). This code and knowledge were used to produce and validate the price of an individual swaption in QuantLib (configuring an initial curve/cube so the NPV matches with internal systems/verification). (Knowledge as in the knowledge of specifics surrounding conventions in how the data is formatted and how that interacts with configuring pricing in order to produce a verified correct price).

Various other packages used in the code: pandas, numpy, QuantLib (as mentioned), math and matplotlib, with the code being produced in python.

7.1 Allowed AI Usage - Course Policy

From the AI Policy course page, paraphrased/translated by me, with reservation for eventual errors. The AI policy can be found at the course page in the original Swedish and this text should not be seen as an exact/binding copy of the original text, but simply my best translation.

Generative AI is allowed to be used if it is used in a way that is:

1. Responsible

2. Ethical
3. Academically honorable and
4. Furthers your learning

Some specific examples of allowed usage were given:

1. Obtaining knowledge: Use AI to find information or to receive understanding of terms, concepts and principles (*"[...]termer, begrepp, och principer."*). Control/make sure that the information is correct, meaning that the sources report what the tool does, ask your adviser, etc!
2. Idea generation: Use AI to get inspiration, structure thoughts or formulate questions. It is, however, always good to first think independently so that the AI tool does not limit your perspectives.
3. Control text/code ("code and text" from the summary in the first part of the original AI policy in Swedish)

In addition, some examples were provided regarding presenting AI usage: (Paraphrased/translated by me)

1. ChatGPT was used to understand the following concepts: [...]
2. Gemini was used to create Image 1. [...]
3. UNWIND was used to debug the R-scripts for analysis of genetic expressions [...]

As such I will now present my ai usage in a similar format, where I have made sure to follow similar use cases.

7.2 AI Usage Report

In the report-specific work, AI (Microsoft Copilot/ChatGPT) was used.

1. AI was used to debug code and odd results, especially QuantLib errors as they are presented in C++ which I don't have knowledge of (Presented in C++ because the python package is a C++ wrapper) (Debugging (Control text/code) use case)

2. AI was used to receive help in understanding QuantLib (as it has very limited python documentation) along with the occasional question regarding other packages (Obtaining knowledge use case)

3. AI was used to receive help in generating figures/tables in the report. Example: help in producing latex code in python to insert the large tables (Image generation use case)

4. AI was used for studying/understanding various theoretical knowledge. Example: Question asked ChatGPT: "In the "Margin requirements for non-centrally cleared derivatives" by the BIS, they are talking about "haircuts" in the context of margin. What is this?" (Obtaining knowledge use case)

5. AI was used for finding good + relevant literature for furthering my understanding/further studies (Obtaining knowledge use case)

6. AI was used to "bounce ideas" regarding both pure theory and regarding interpreting the results/the discussion. When doing this an effort was made to always think through the theory/interpretation deeply and well before. The resulting understanding/interpretation in the text was checked and discussed with my industrial adviser Coën Olofsson. (Obtaining knowledge/Idea generation use case)

7. AI was used to check a couple of the more convoluted mathematical formulas for mistakes after they were written in LaTeX (Control text/code use case)

7.2.1 AI Use/Report Approval by Adviser

The actual AI usage was checked against the AI policy by my adviser Kirthana Rajasekar, along with this AI usage report, both of which were approved by her. (Both the AI use and the AI disclaimer were OK/approved by her).

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